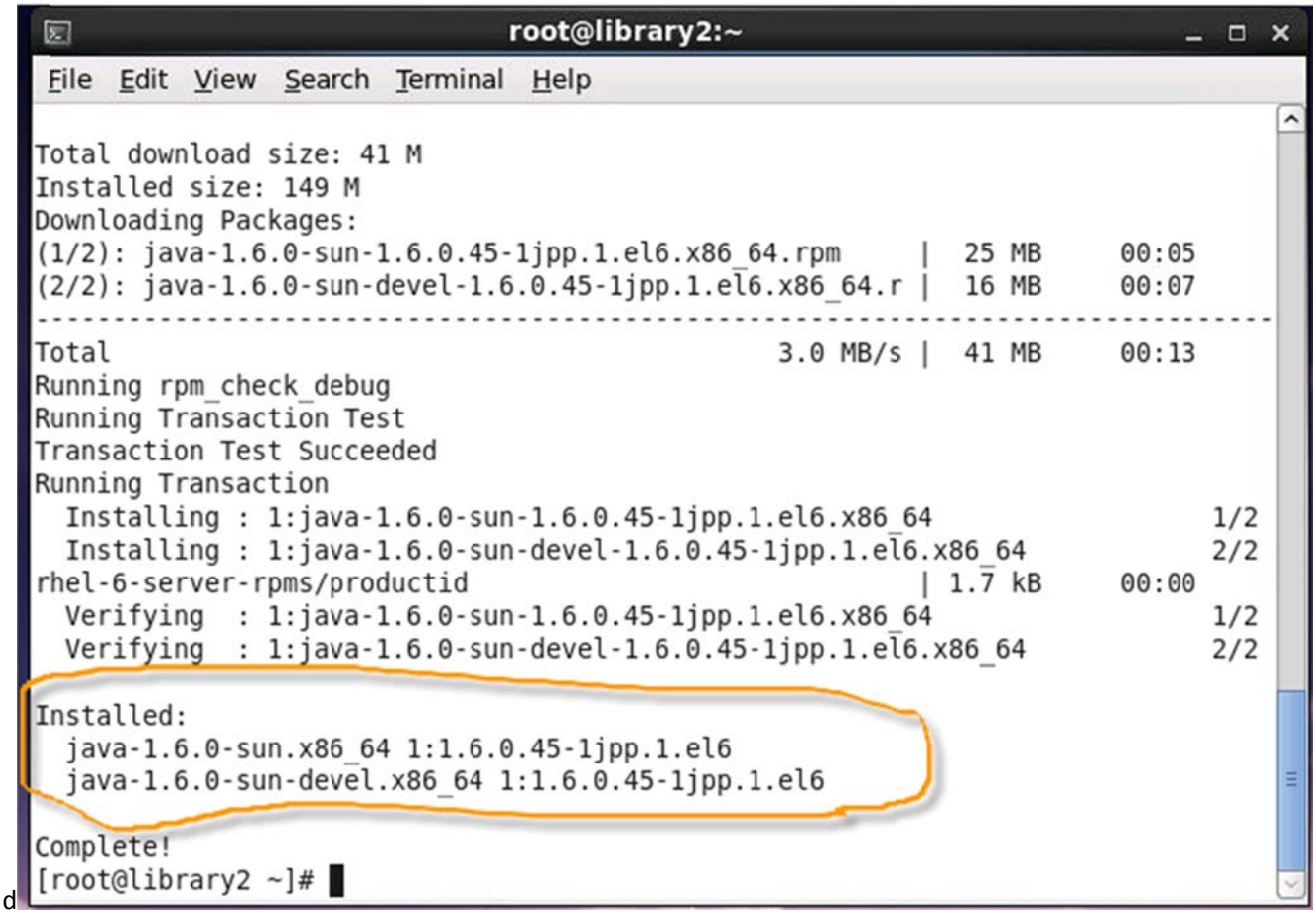
DSpace Installation on Redhat Linux server Step by step

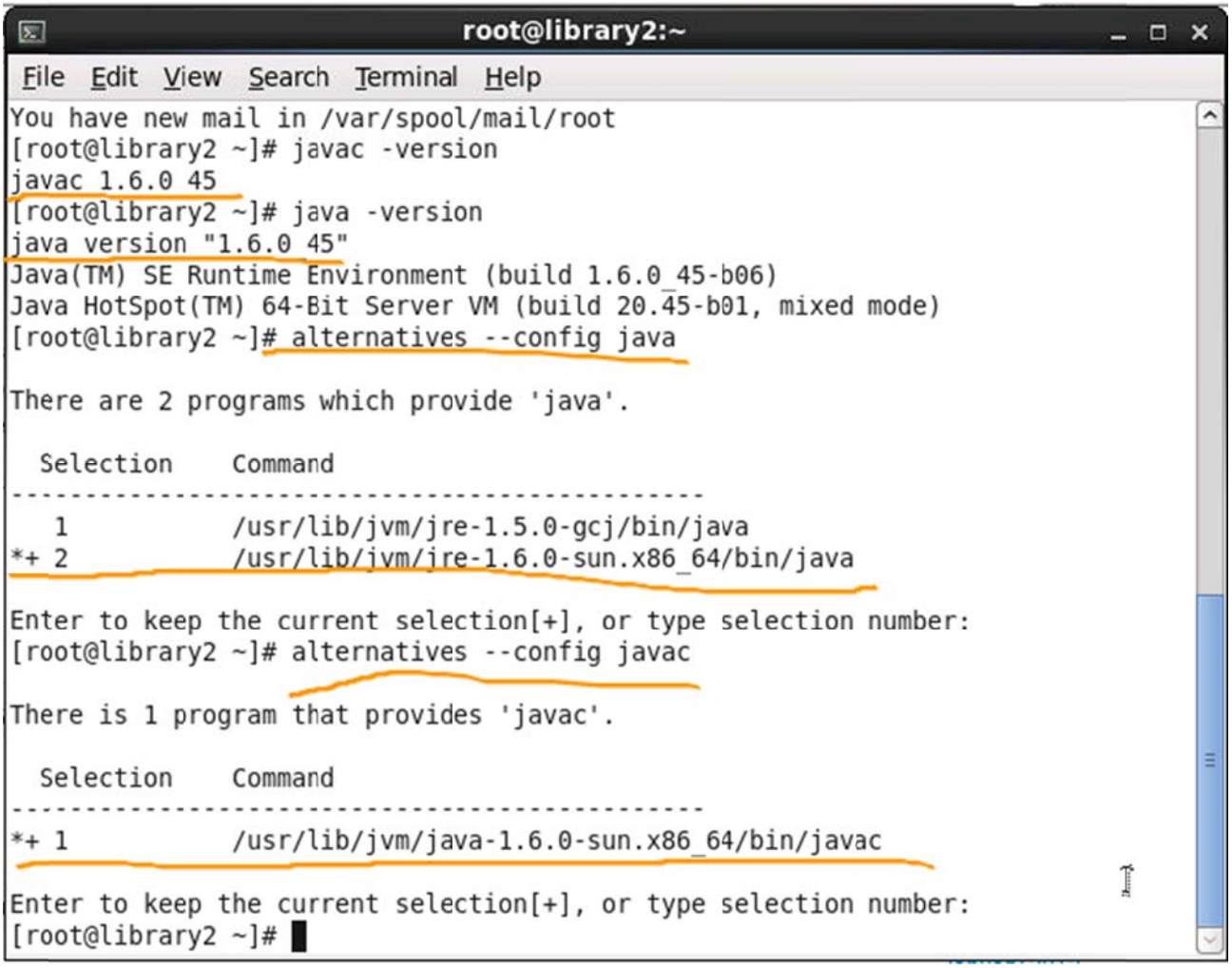
Andy Tang
Systems and Electronic Resources Librarian
Kilgore College

1. Go to the Redhat linux server to check whether the java is installed.

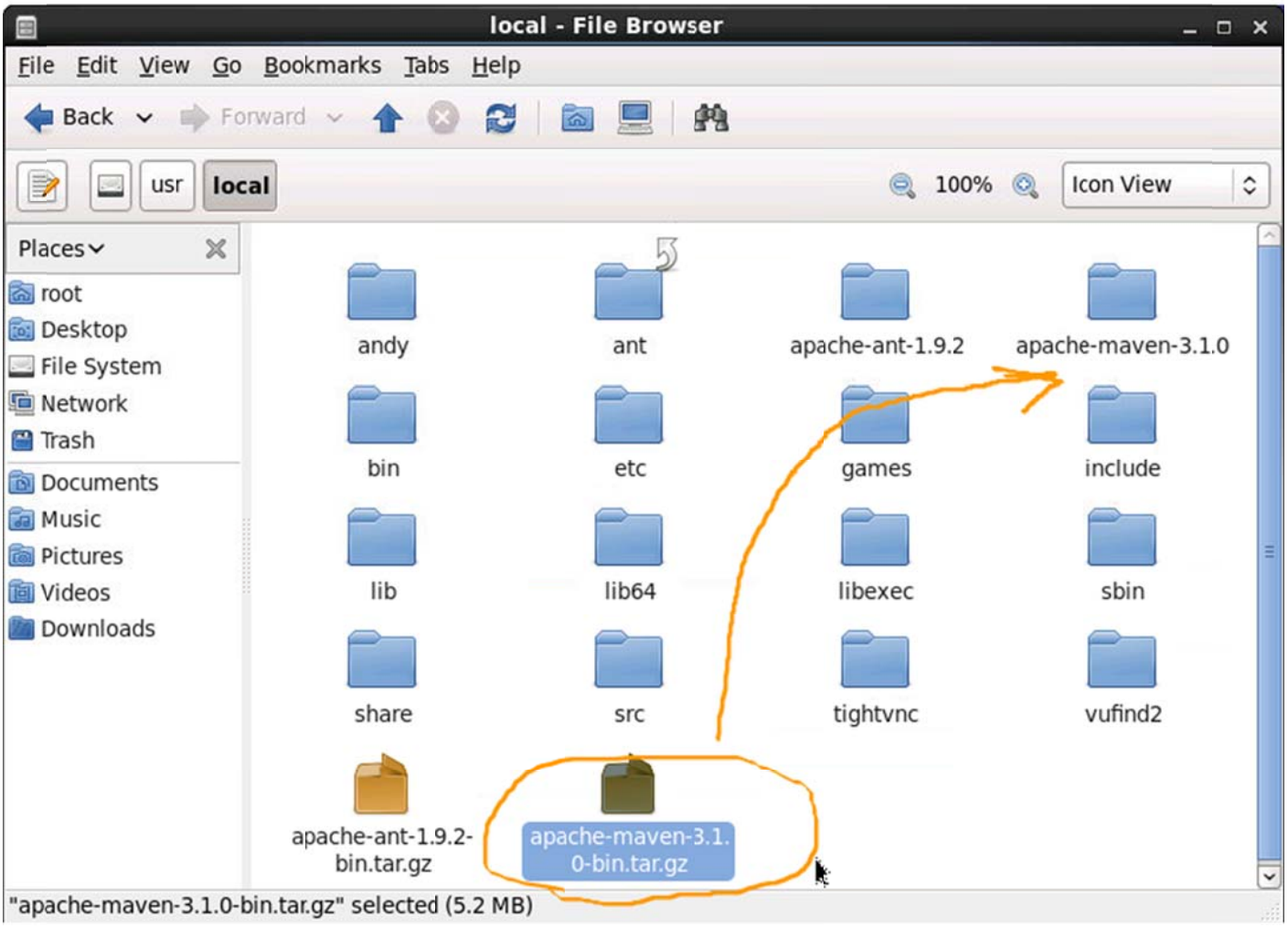


d

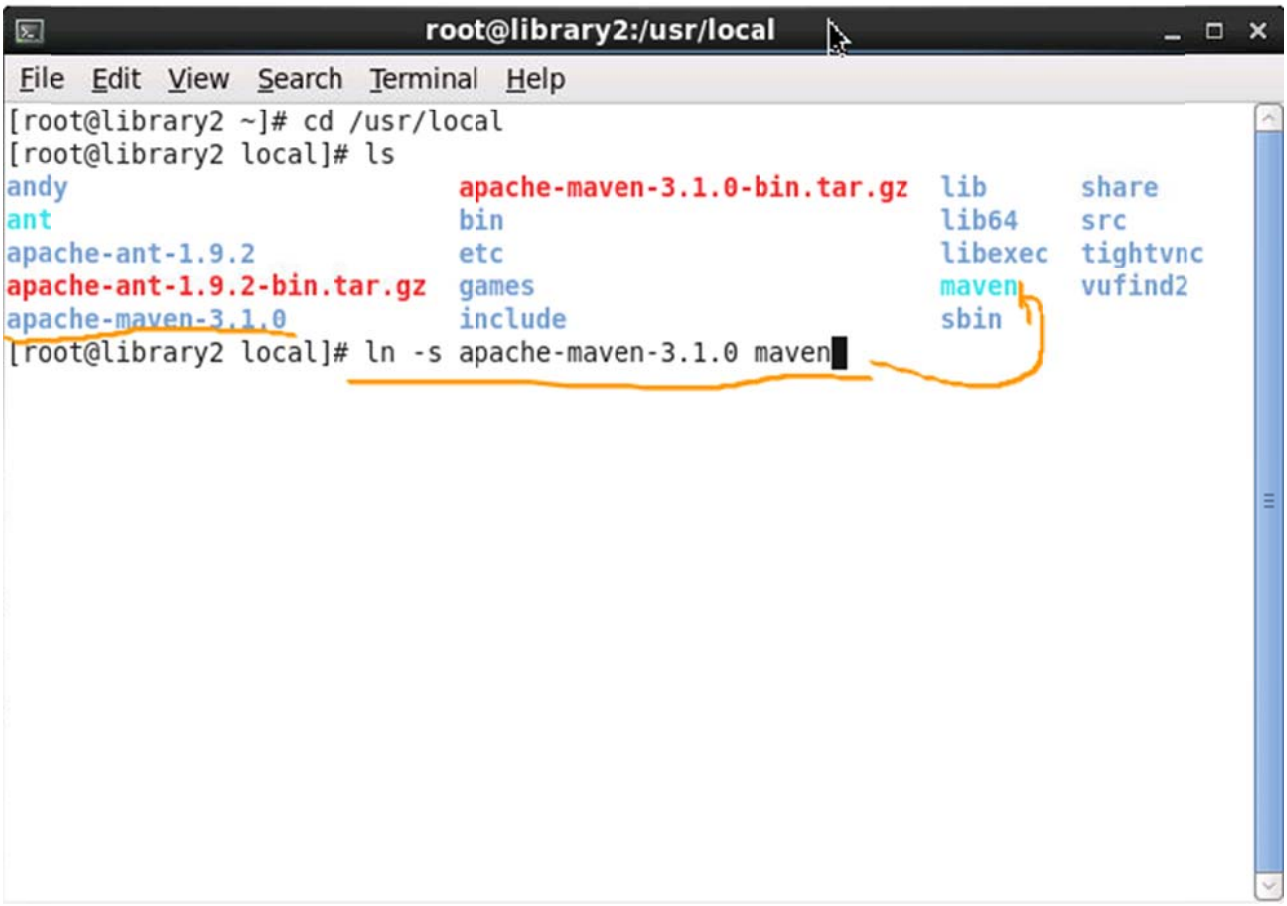
2. By using alternative command, the correct version of java is used as the default.



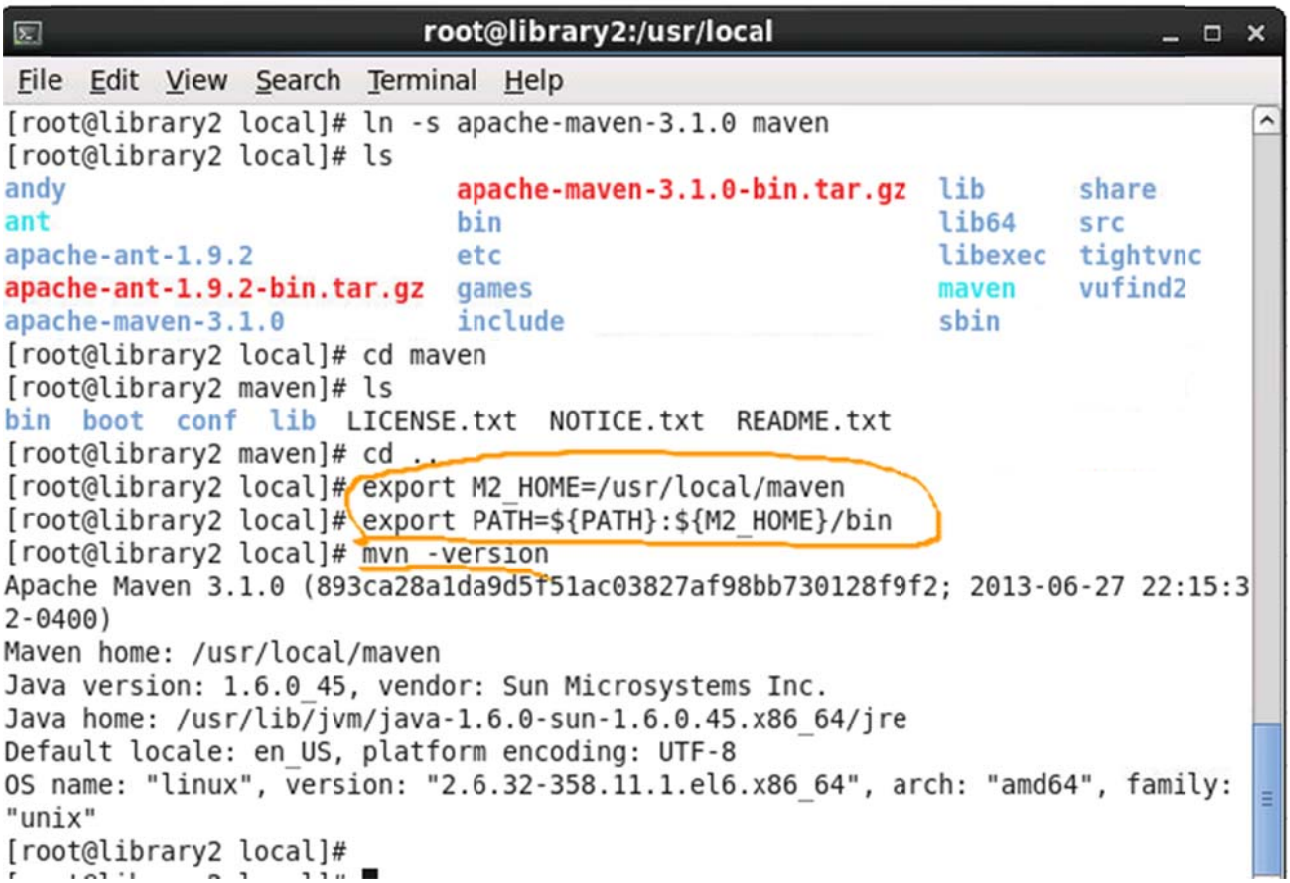
3. Download Apache maven and uncompress it.



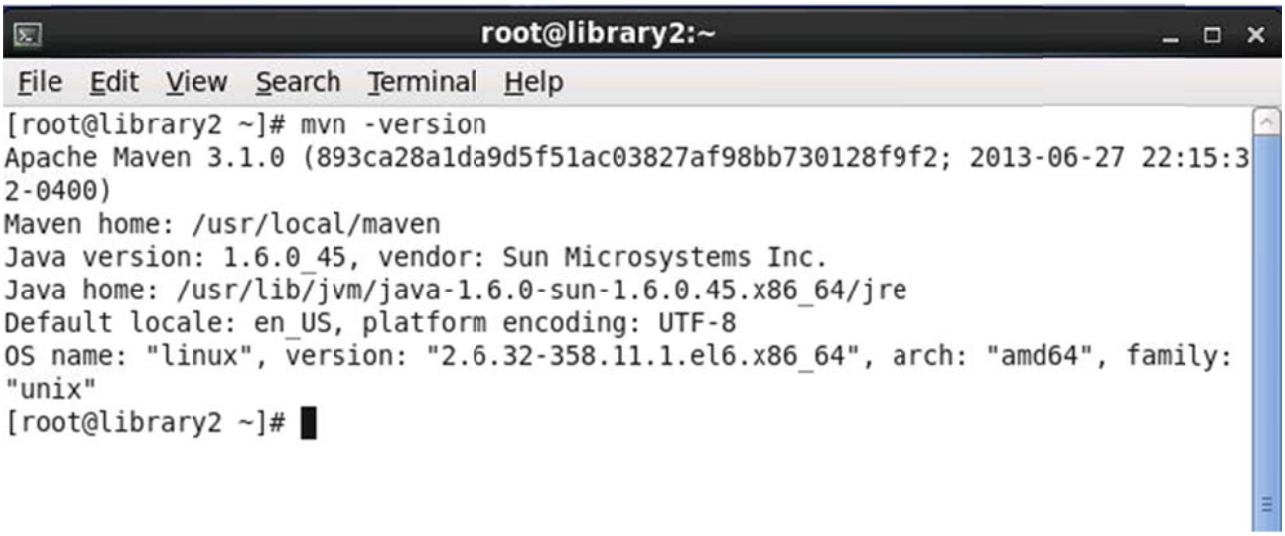
4. Create symbolic link for Apache maven folder.



5. Setup environment for Apache Maven



6. Confirm that Apache Maven is installed successful.



7. Check the verison of Apache Ant.

```
[root@library2 ~]# ant -version
Apache Ant(TM) version 1.9.2 compiled on July 8 2013
[root@library2 ~]#
```

8. Check the version of PostGreSql.

```
[root@library2 ~]# rpm -q postgresql
postgresql-8.4.13-1.el6_3.x86_64
[root@library2 ~]# rpm -q postgresql-server
postgresql-server-8.4.13-1.el6_3.x86_64
[root@library2 ~]#
```

9. Restart PostGreSQL service

```
[root@library2 ~]# service postgresql restart
Stopping postgresql service:                               [  OK  ]

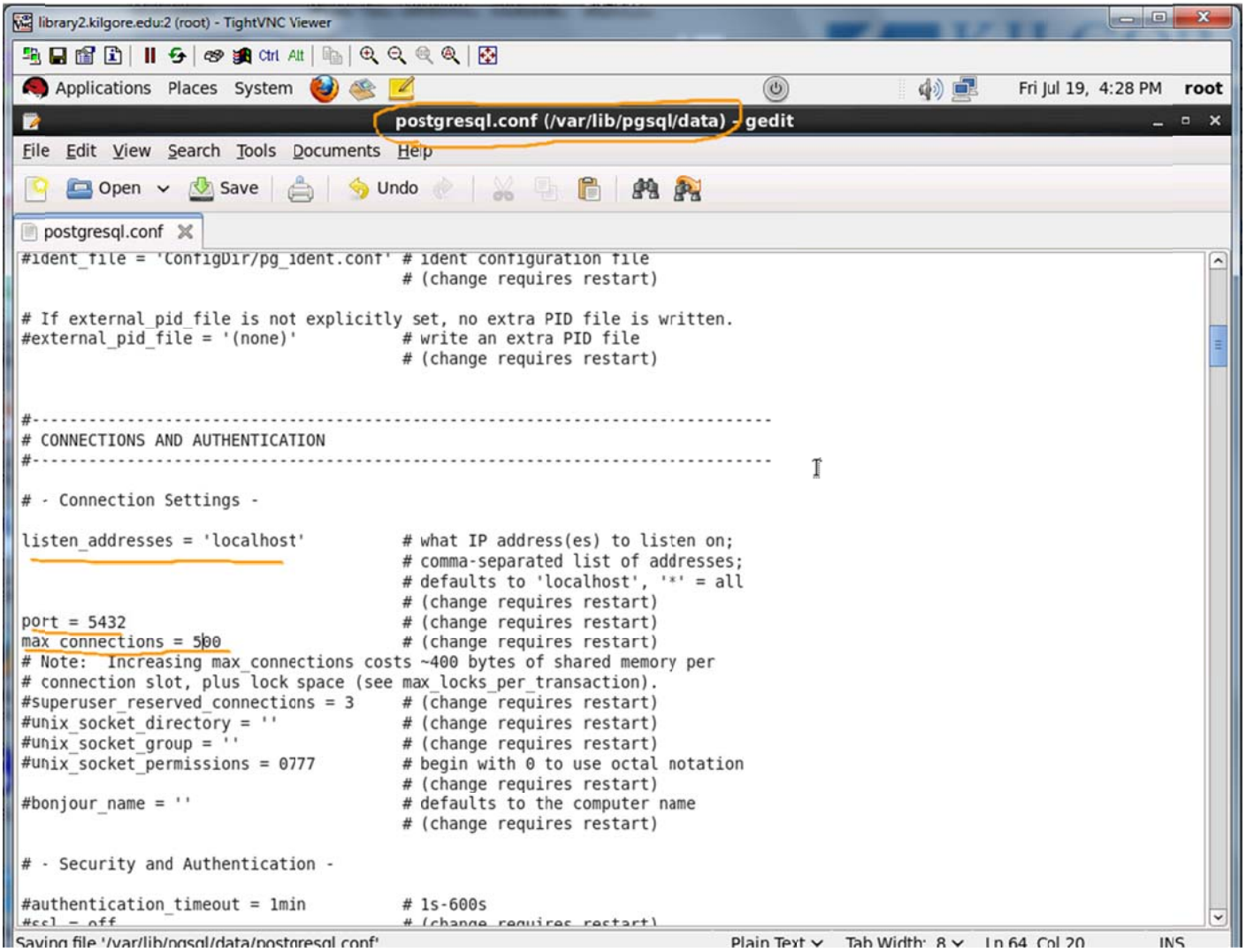
/var/lib/pgsql/data is missing. Use "service postgresql initdb" to initialize th
e cluster first.
                                                           [FAILED]
[root@library2 ~]#
```

10. Run Service PostGreSQL initdb command to initialize the cluster.

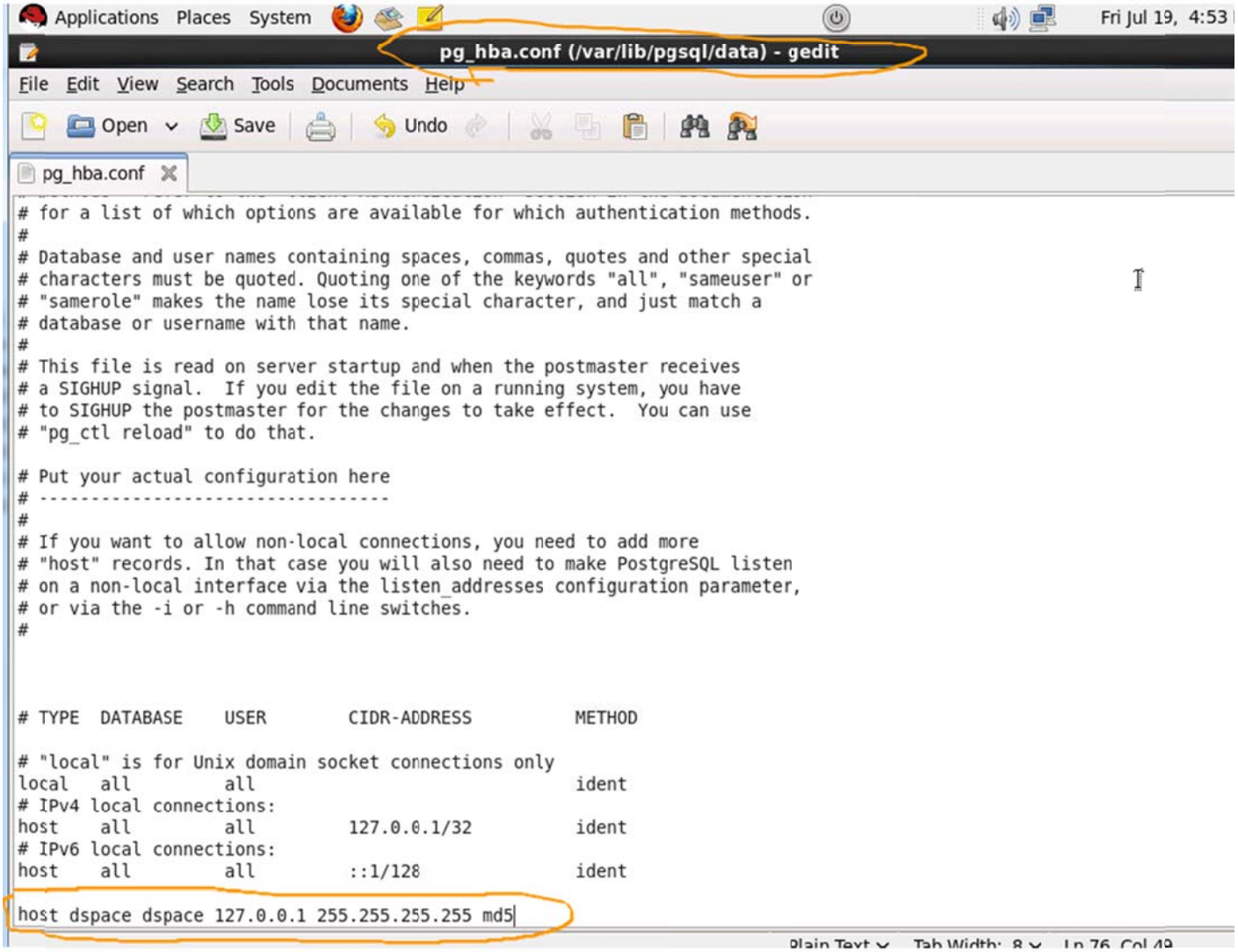
```
postgres: unrecognized service
[root@library2 ~]# service postgresql restart
Stopping postgresql service:                               [  OK  ]

/var/lib/pgsql/data is missing. Use "service postgresql initdb" to initialize th
e cluster first.
                                                           [FAILED]
[root@library2 ~]# service postgresql initdb
Initializing database:                                     [  OK  ]
[root@library2 ~]#
```

11. Check the listen_address, port, and max_connection parameter in the file of pstgresql.conf/.



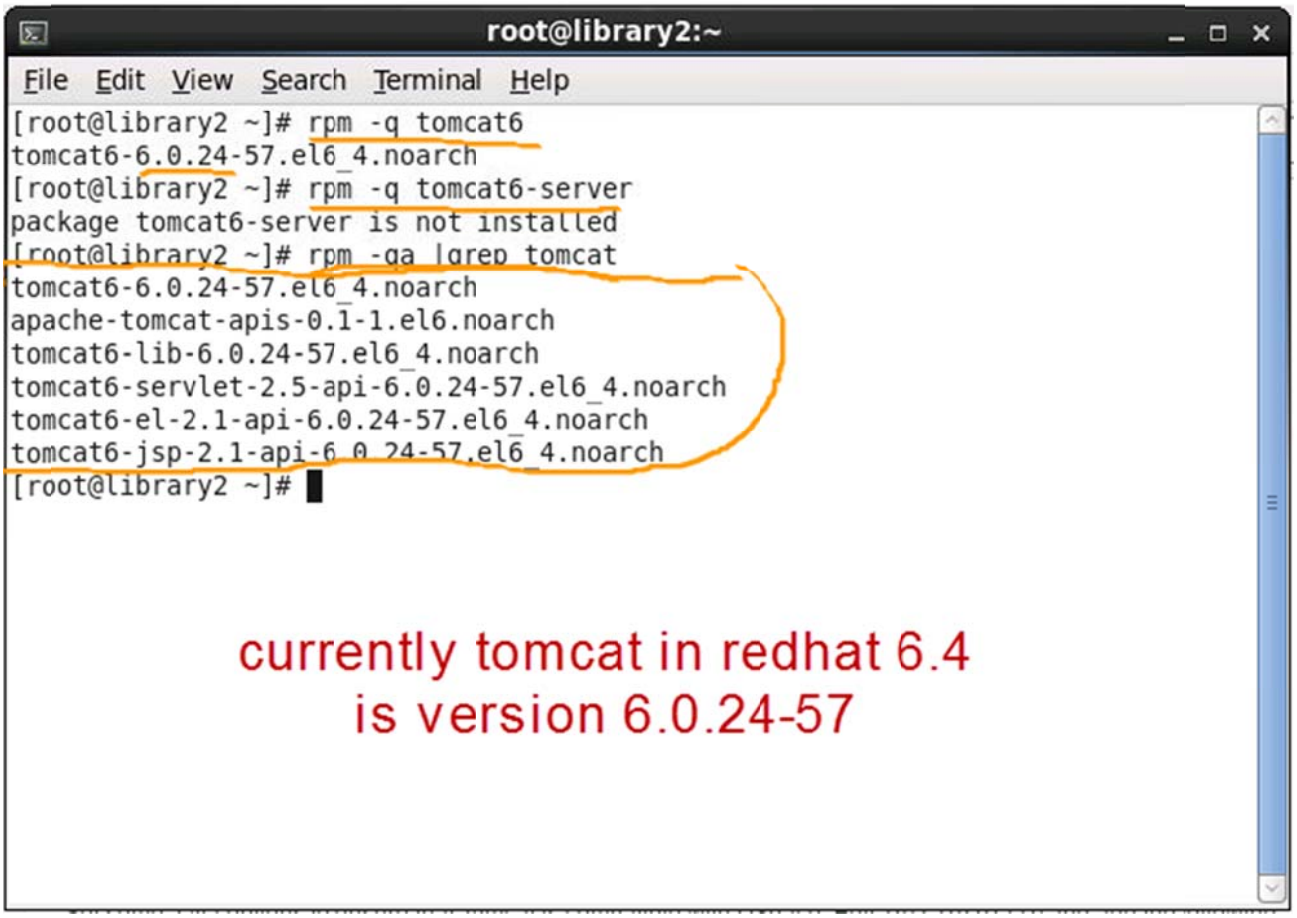
12. Who can login and what IP range is allowed to access.



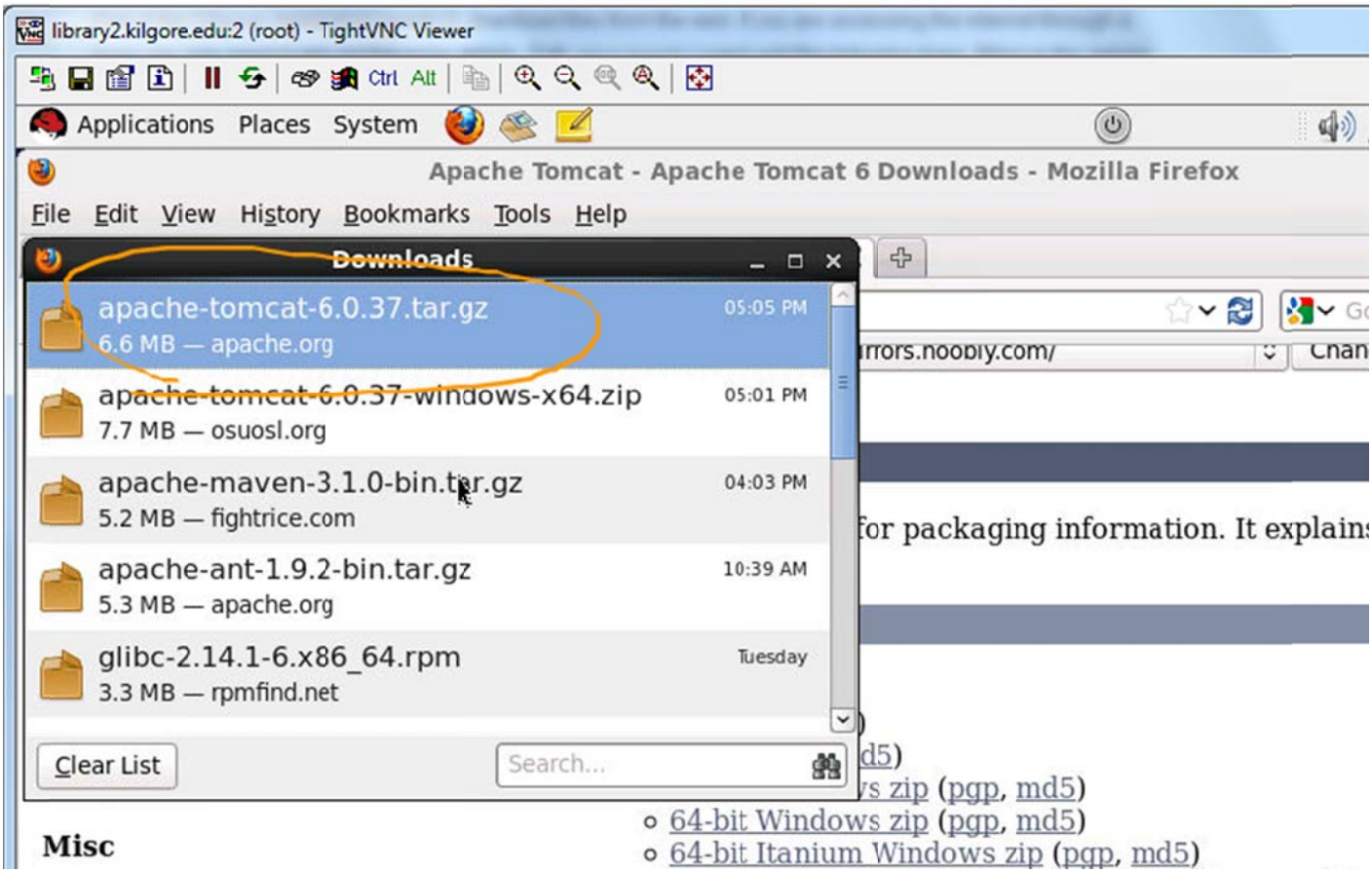
13. Restart PostGreSQL database service.



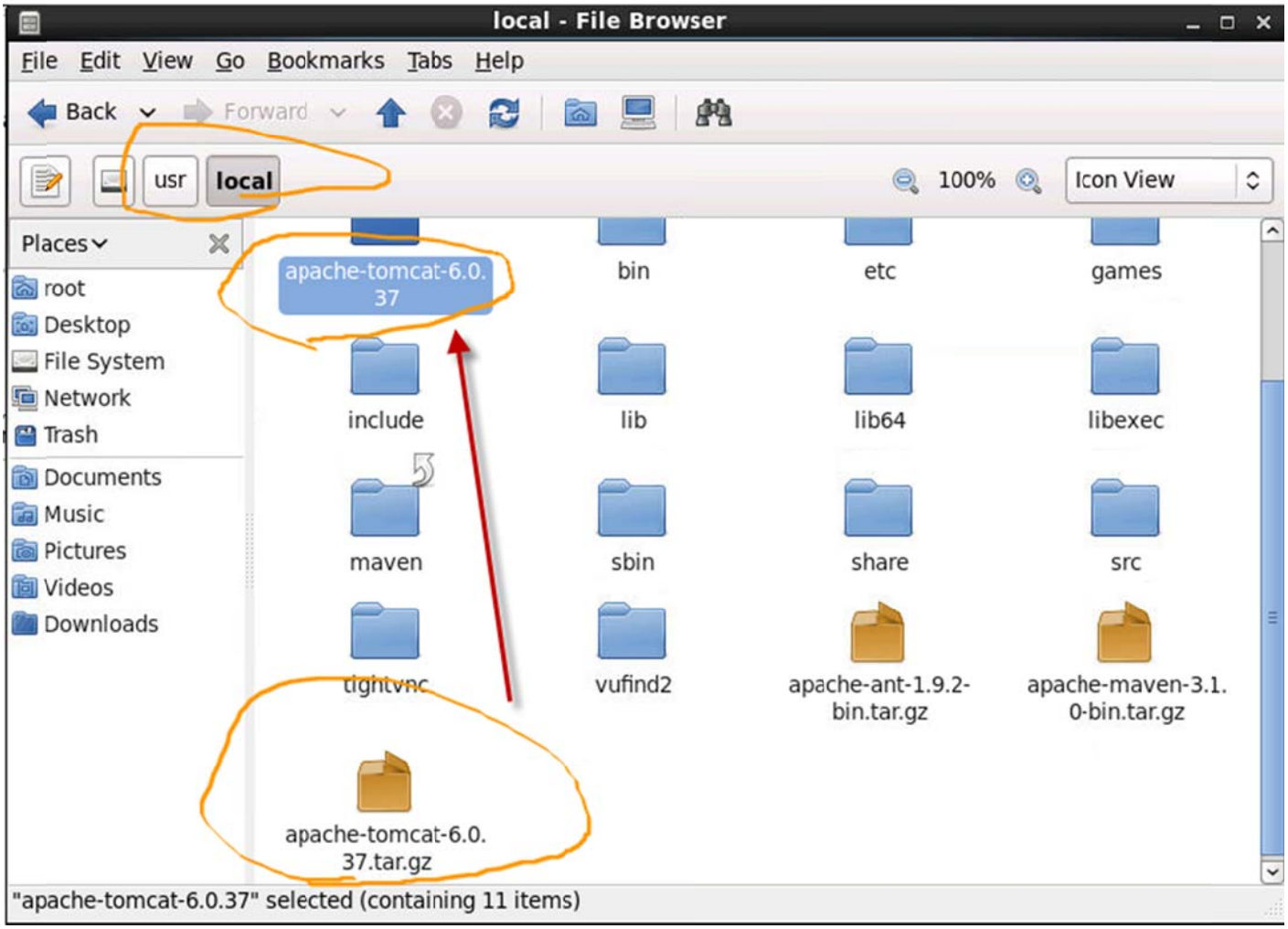
14. Check the version of Tomcat.



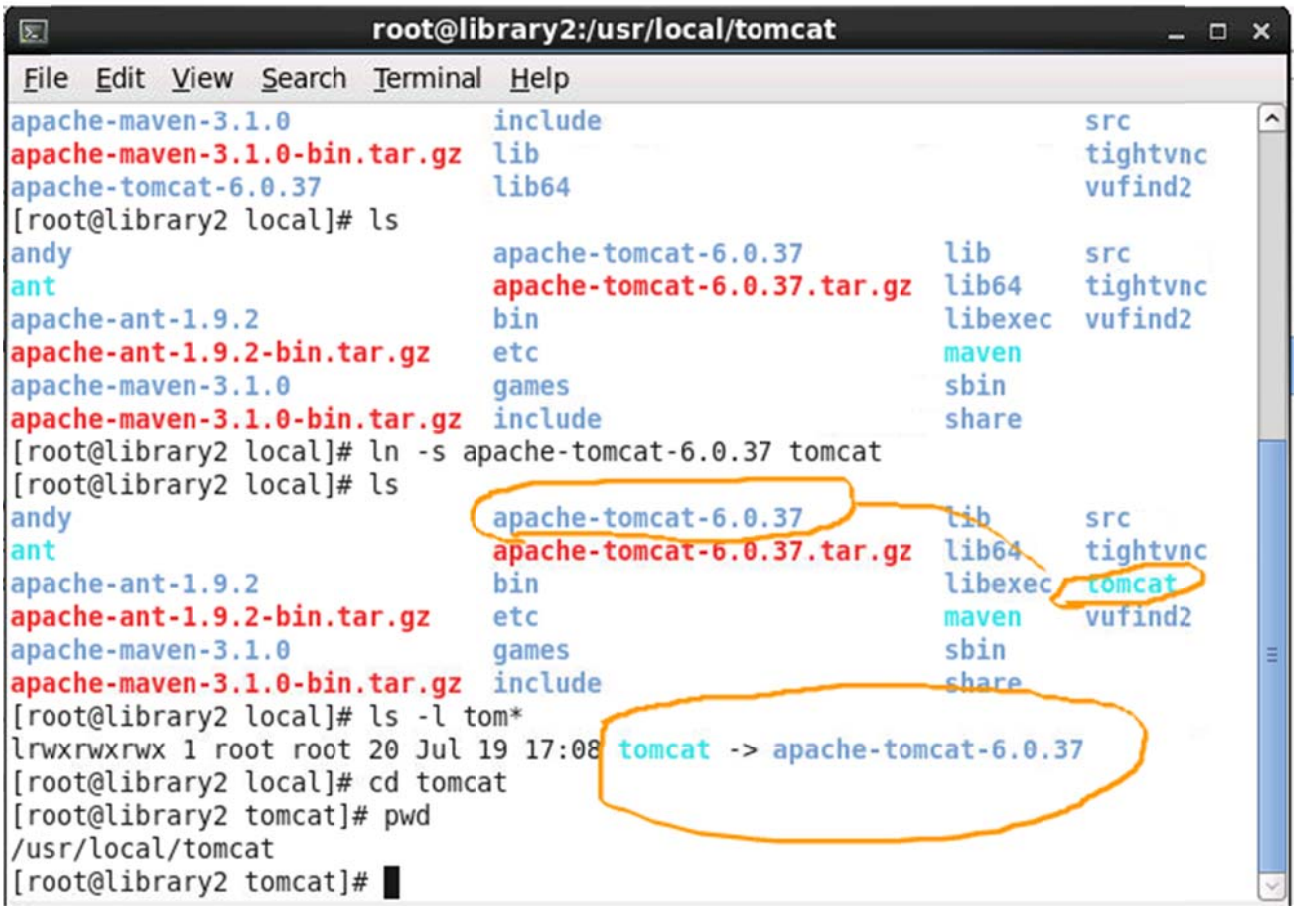
15. We determined to use Apache tomcat 6.0.37



16. Download and extract the apache-tomcat-6.0.37.tar.gz



17. Create symbolic link for folder apache-tomcat-6.0.37



18. Allocate user and usergroup authority for folder apache-tomcat-6.0.37

```
[root@library2 local]# cd tomcat
[root@library2 tomcat]# pwd
/usr/local/tomcat
[root@library2 tomcat]# cd /usr/local
[root@library2 local]# chown -R dspace:dspace apache-tomcat-6.0.37
[root@library2 local]# ls -l apa*
-rw-r--r-- 1 root   root   5512093 Jul 19 10:39 apache-ant-1.9.2-bin.tar.gz
-rw-r--r-- 1 root   root   5439122 Jul 19 16:03 apache-maven-3.1.0-bin.tar.gz
-rw-r--r-- 1 root   root   6885442 Jul 19 17:05 apache-tomcat-6.0.37.tar.gz

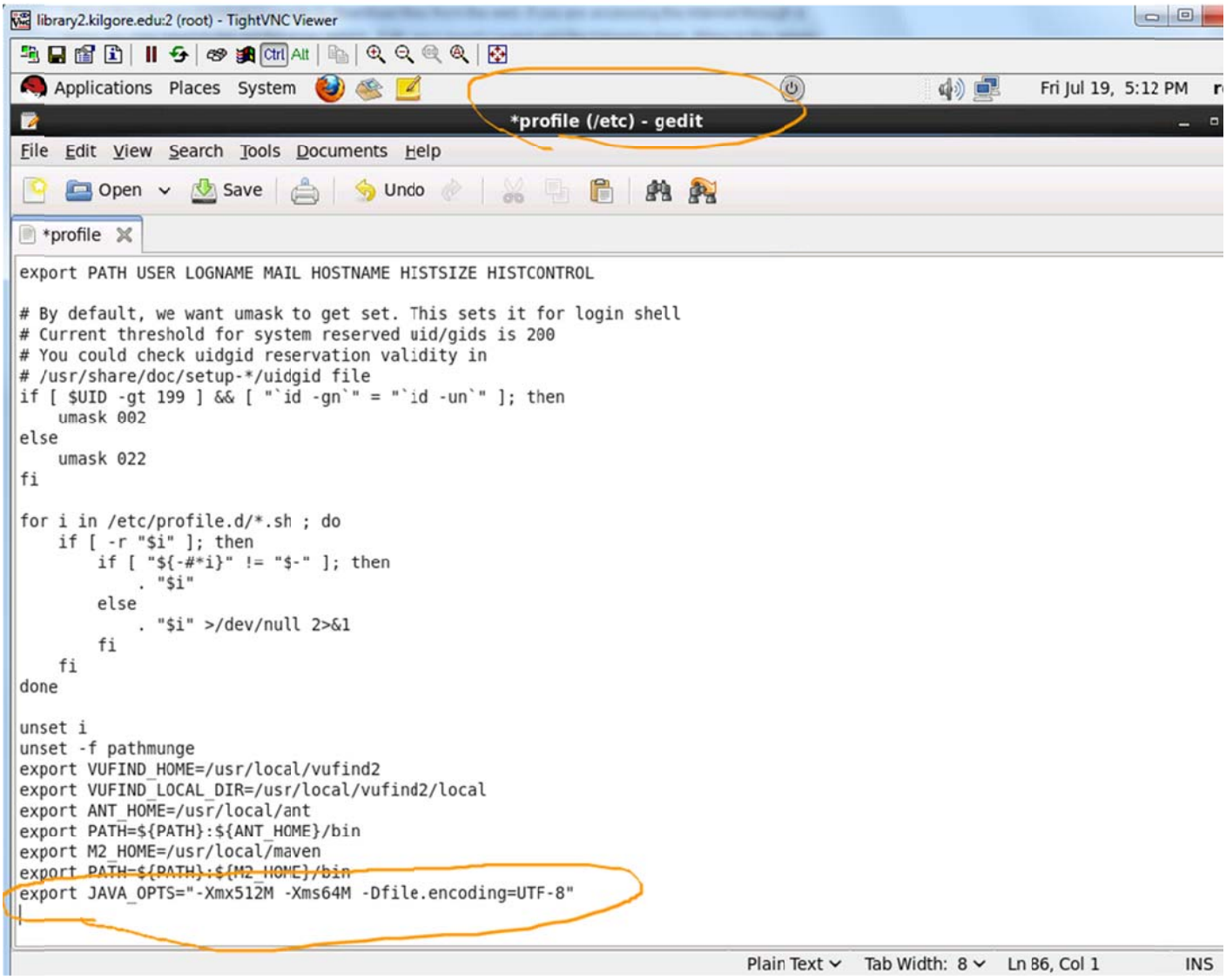
apache-ant-1.9.2:
```

19. Check whether the allocation of authority is successful or not.

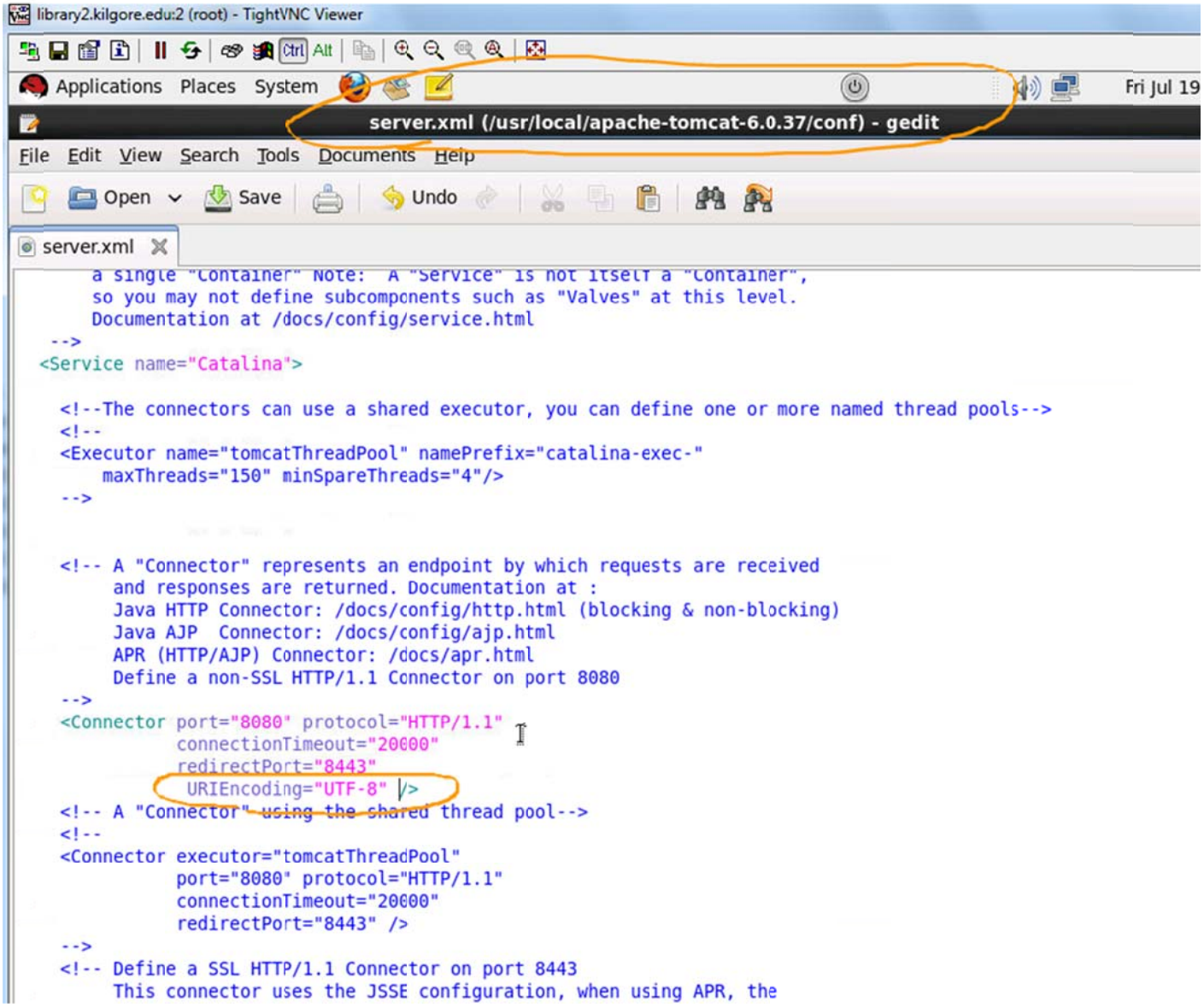
```
root@library2:/usr/local
File  Edit  View  Search  Terminal  Help
-rw-r--r-- 1 root root 11358 Jun 27 22:12 LICENSE.txt
-rw-r--r-- 1 root root  1112 Jun 27 22:12 NOTICE.txt
-rw-r--r-- 1 root root  2513 Jun 27 22:12 README.txt

apache-tomcat-6.0.37:
total 100
drwxr-xr-x 2 dspace dspace  4096 Jul 19 17:06 bin
drwxr-xr-x 2 dspace dspace  4096 Apr 29 05:36 conf
drwxr-xr-x 2 dspace dspace  4096 Jul 19 17:06 lib
-rw-r--r-- 1 dspace dspace 37951 Apr 29 05:36 LICENSE
drwxr-xr-x 2 dspace dspace  4096 Apr 29 05:36 logs
-rw-r--r-- 1 dspace dspace   558 Apr 29 05:36 NOTICE
-rw-r--r-- 1 dspace dspace  9188 Apr 29 05:36 RELEASE-NOTES
-rw-r--r-- 1 dspace dspace 16287 Apr 29 05:36 RUNNING.txt
drwxr-xr-x 2 dspace dspace  4096 Jul 19 17:06 temp
drwxr-xr-x 7 dspace dspace  4096 Apr 29 05:36 webapps
drwxr-xr-x 2 dspace dspace  4096 Apr 29 05:36 work
[root@library2 local]#
```

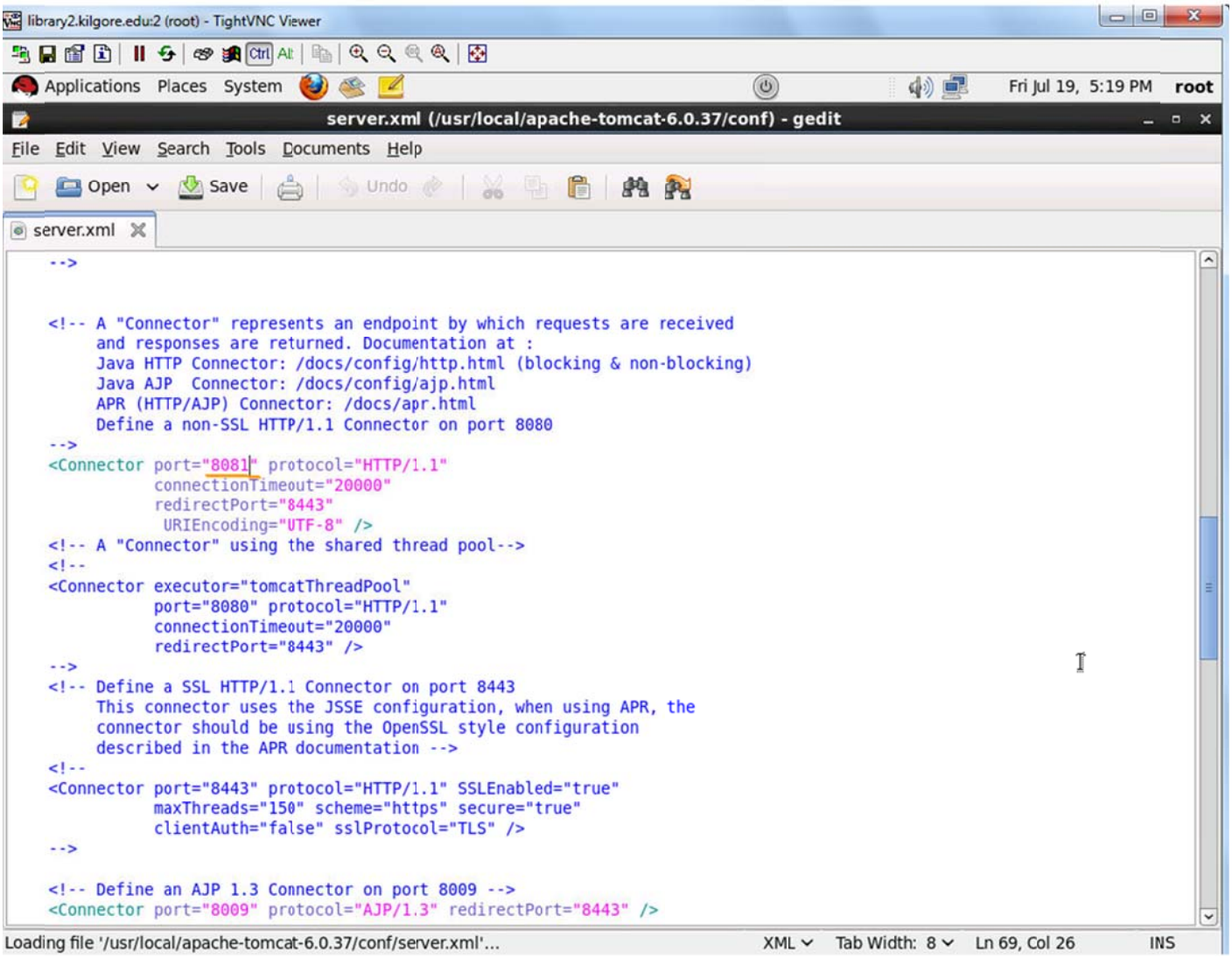
20. Setup java_opts value in /etc/profile



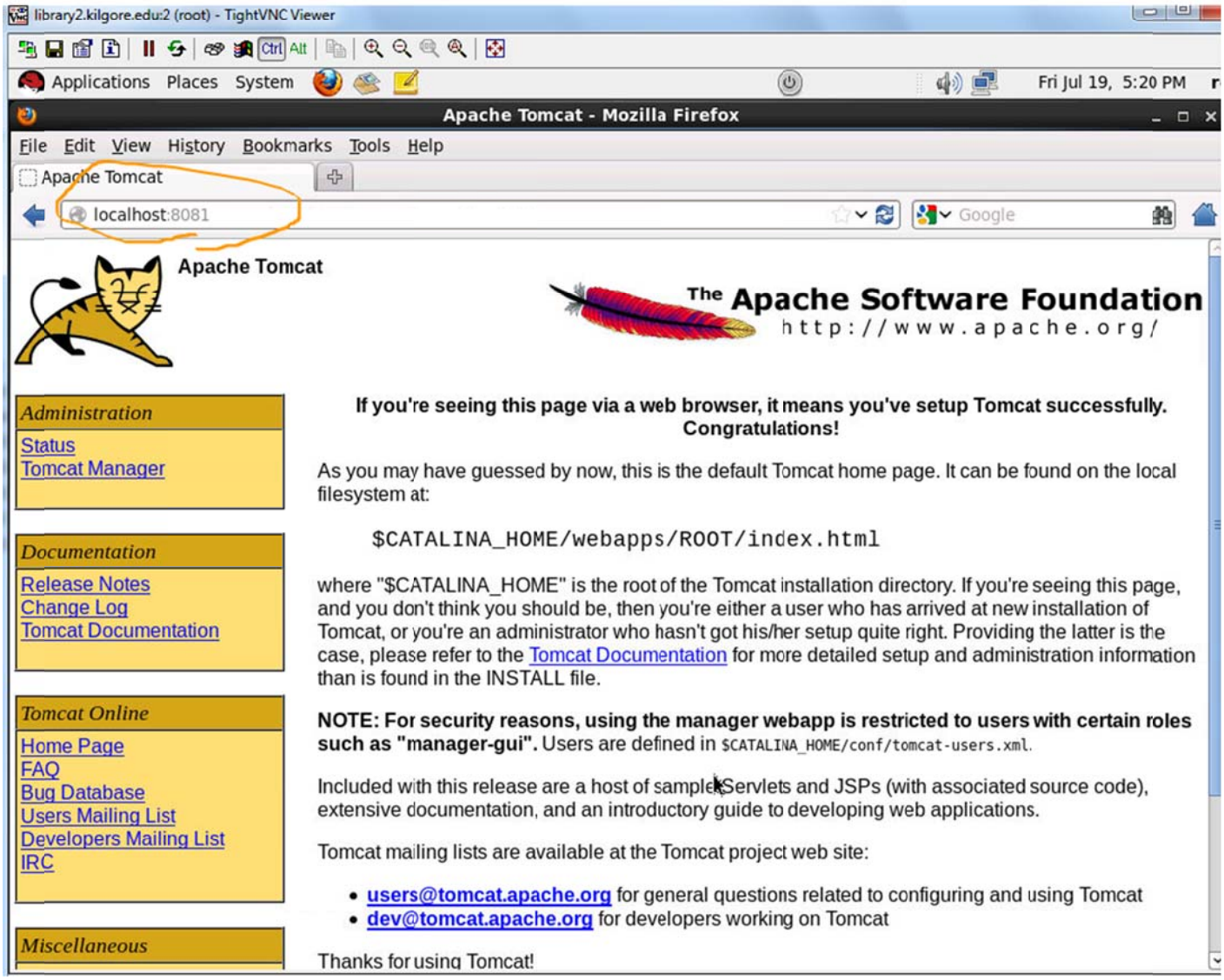
21. Confirm that URLEncoding is UTF-8 in tomcat server.xml file



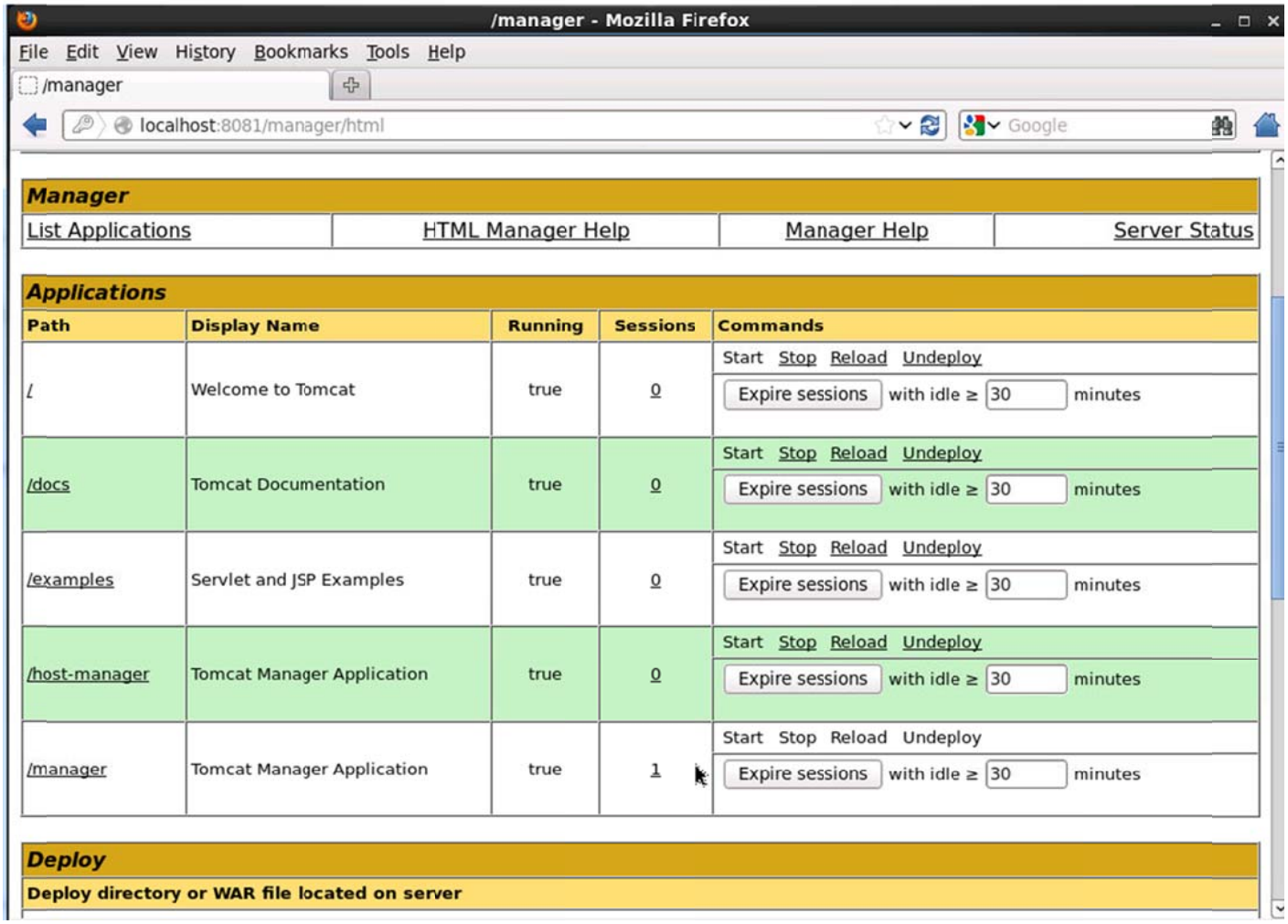
22. Change the connector port to 8081



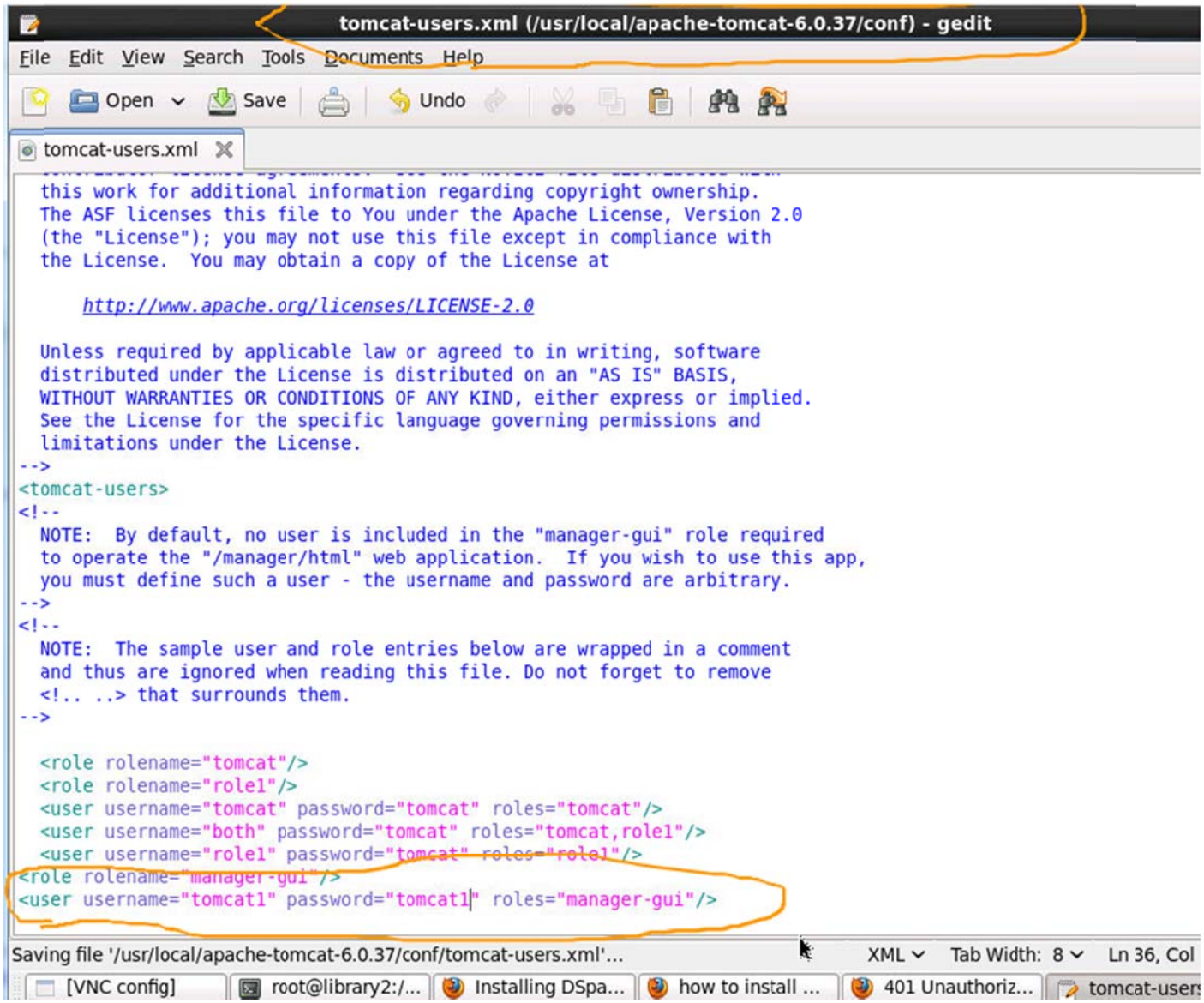
23. Apache Tomcat will use 8081 port



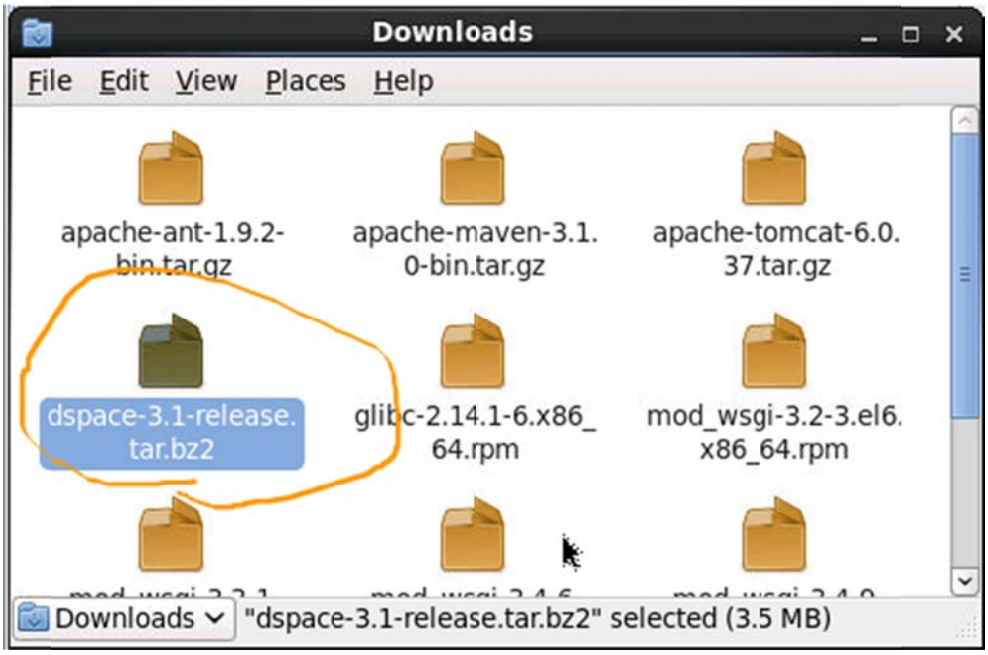
24. It has no problem to login manager interface of Apache Tomcat



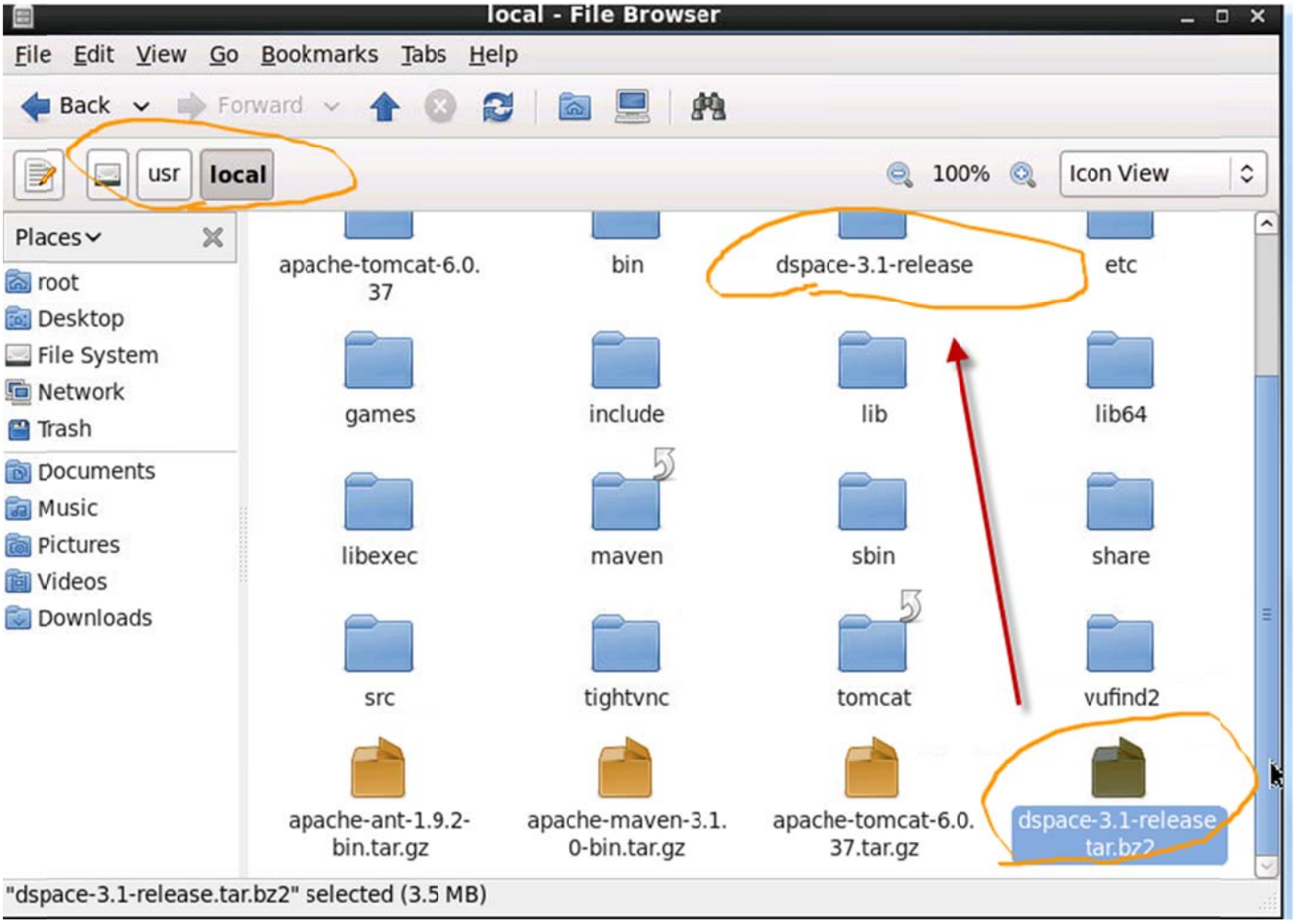
25. Configure the username and password to login manager interface



26. Download installation package of Dspace



27. Uncompress the downloaded installation file dspace-3.1-release.tar.bz2

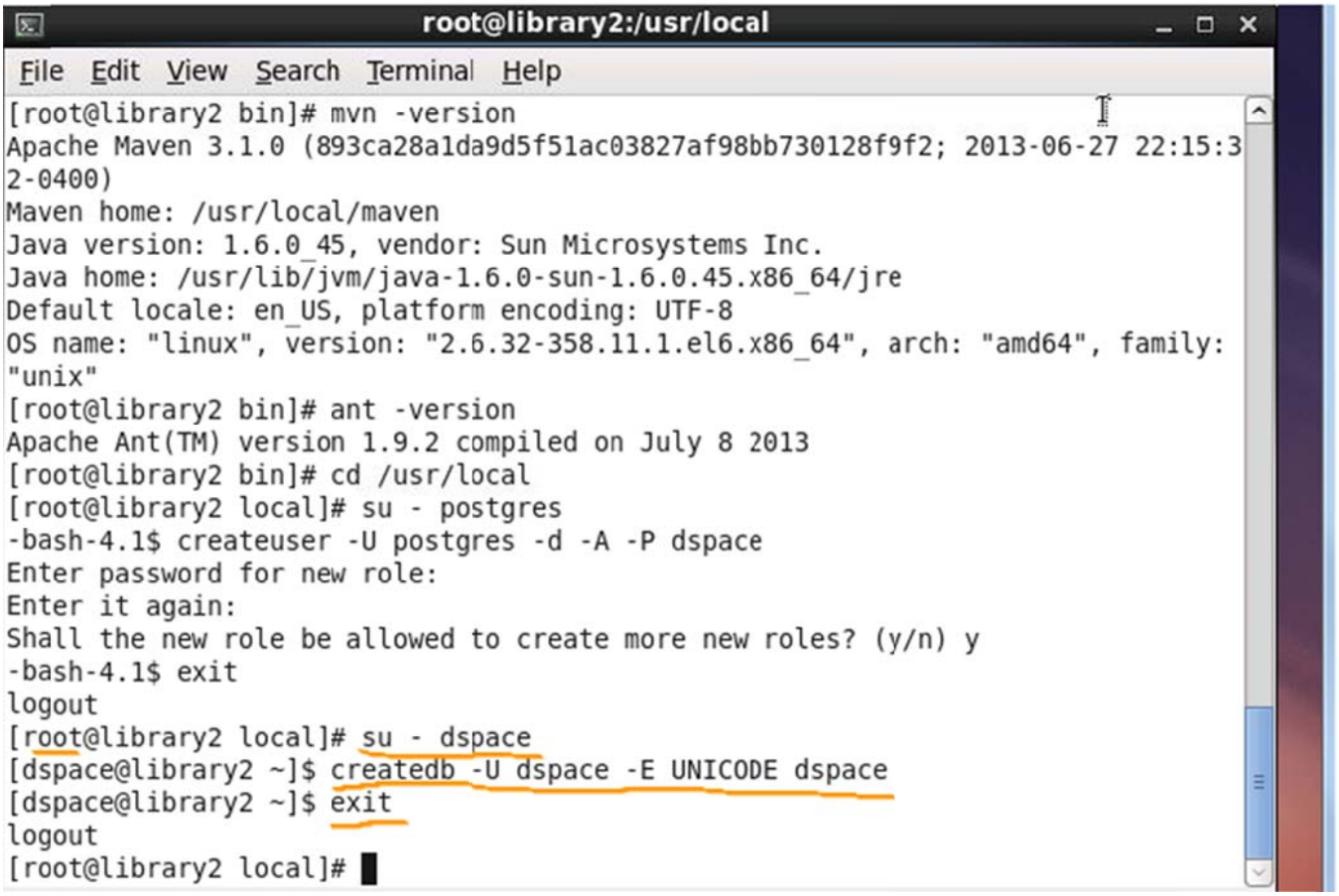


28. Create user for PostGreSQL database

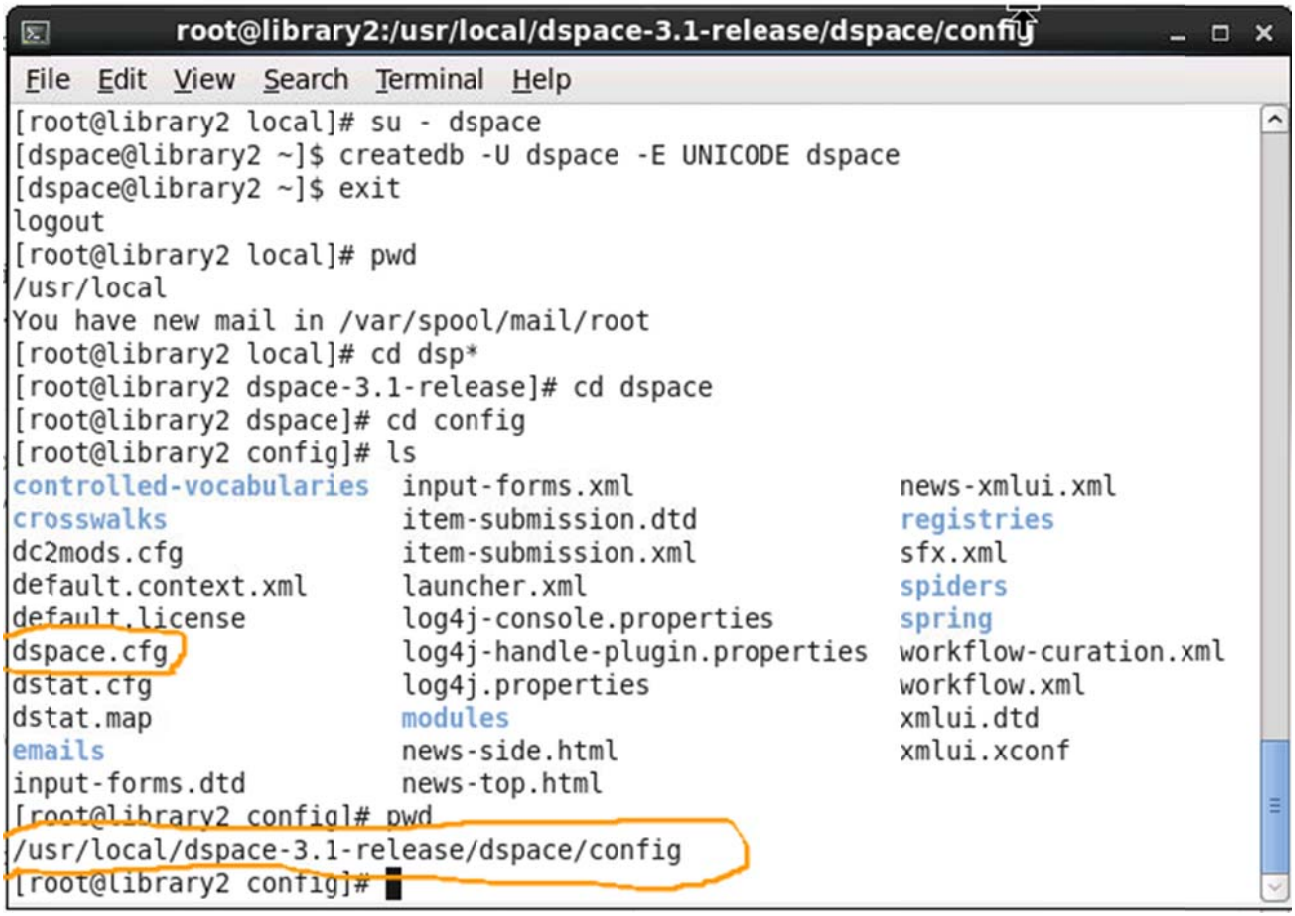
```
Using CATALINA_BASE:   /usr/local/tomcat
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
[root@library2 bin]# mvn -version
Apache Maven 3.1.0 (893ca28a1da9d5f51ac03827af98bb730128f9f2; 2013-06-27 22:15:3
2-0400)
Maven home: /usr/local/maven
Java version: 1.6.0_45, vendor: Sun Microsystems Inc.
Java home: /usr/lib/jvm/java-1.6.0-sun-1.6.0.45.x86_64/jre
Default locale: en_US, platform encoding: UTF-8
OS name: "linux", version: "2.6.32-358.11.1.el6.x86_64", arch: "amd64", family:
"unix"
[root@library2 bin]# ant -version
Apache Ant(TM) version 1.9.2 compiled on July 8 2013
[root@library2 bin]# cd /usr/local
[root@library2 local]# su - postgres
-bash-4.1$ createuser -U postgres -d -A -P dspace
Enter password for new role:
Enter it again:
Shall the new role be allowed to create more new roles? (y/n) y
-bash-4.1$
```



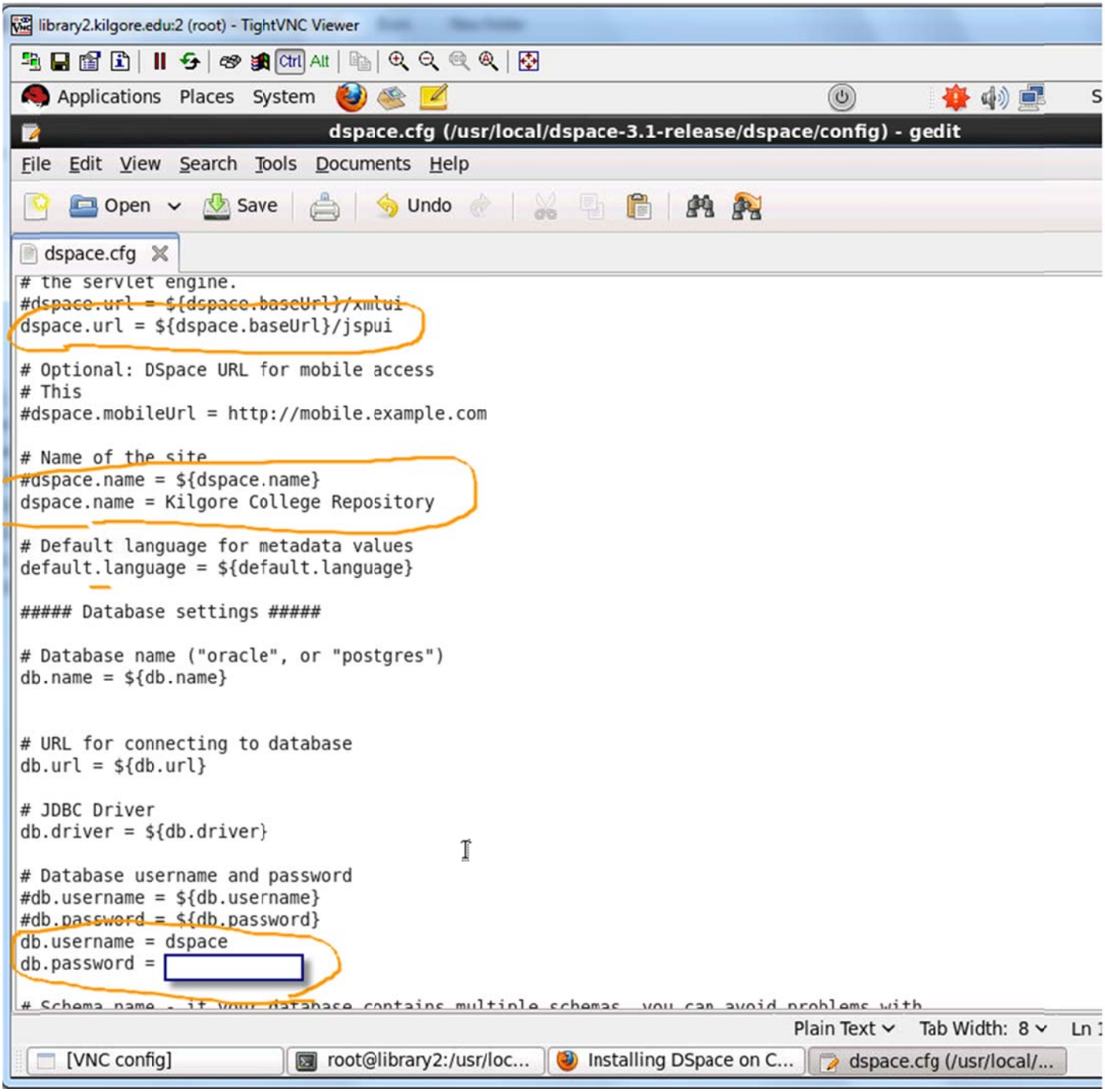
29. Login by using dspace and create dspace database



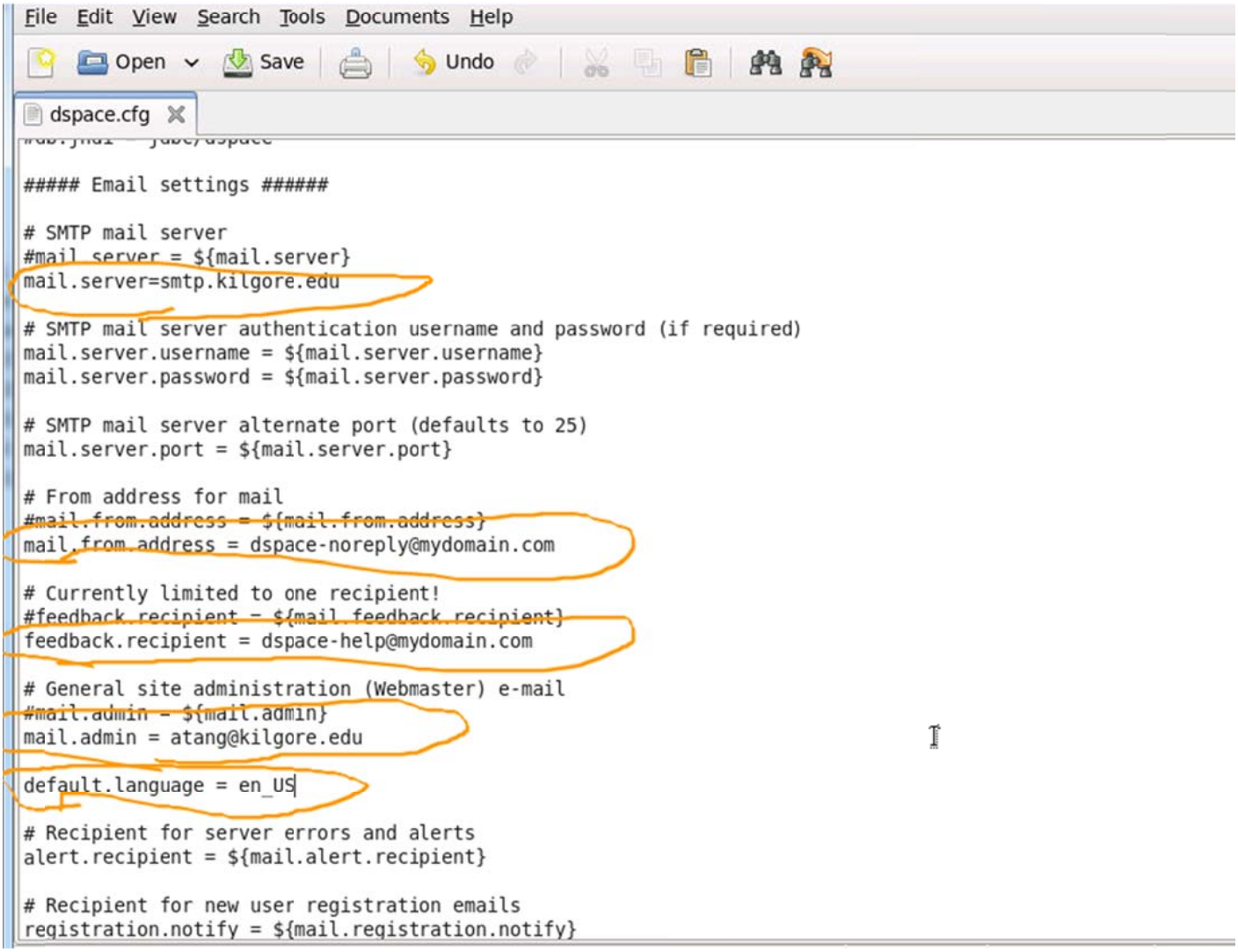
30. Locate dspace.cfg file in the folder /usr/local/dspace-3.1-release/dspace/config



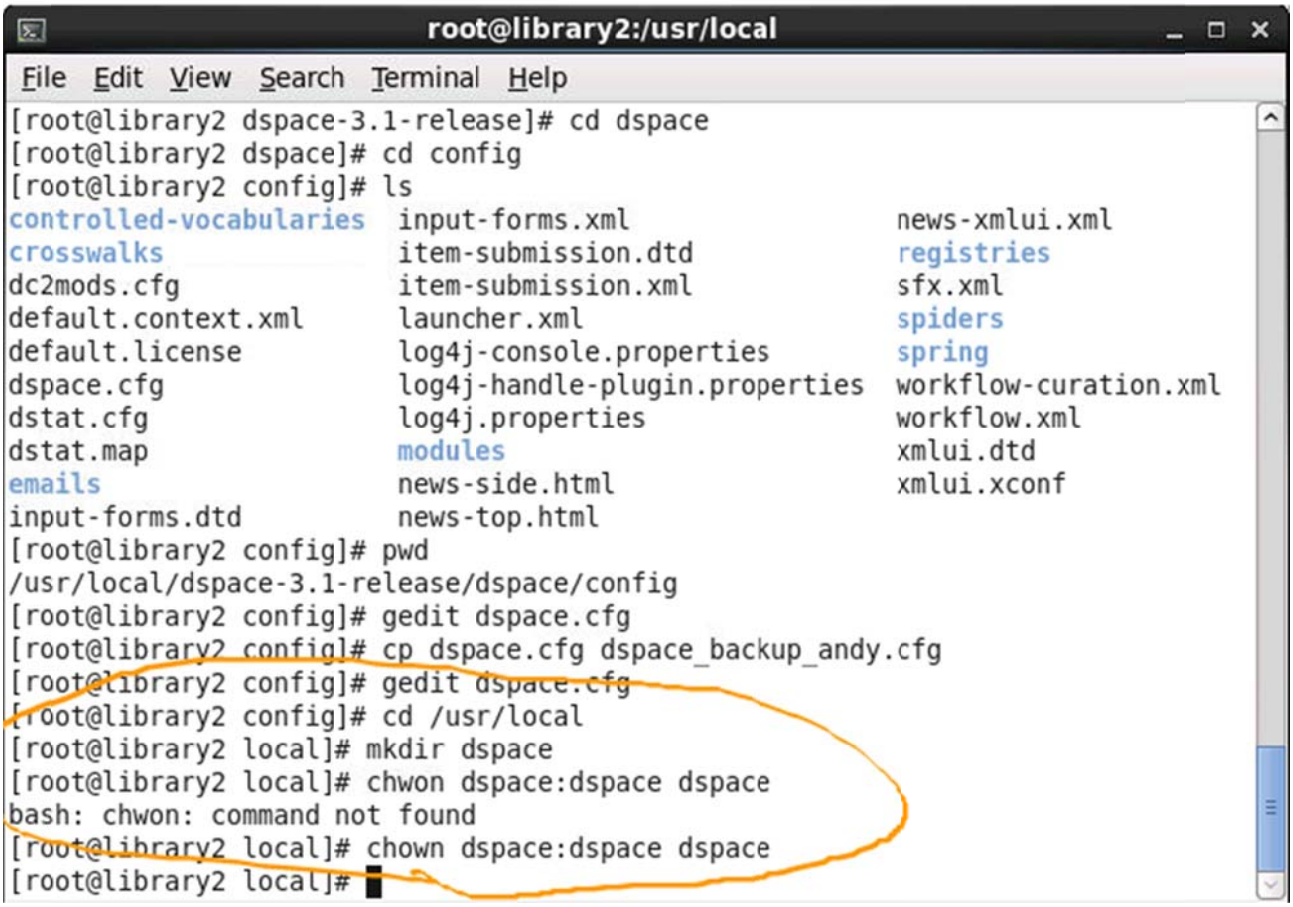
31. Modify the dspace.cfg file



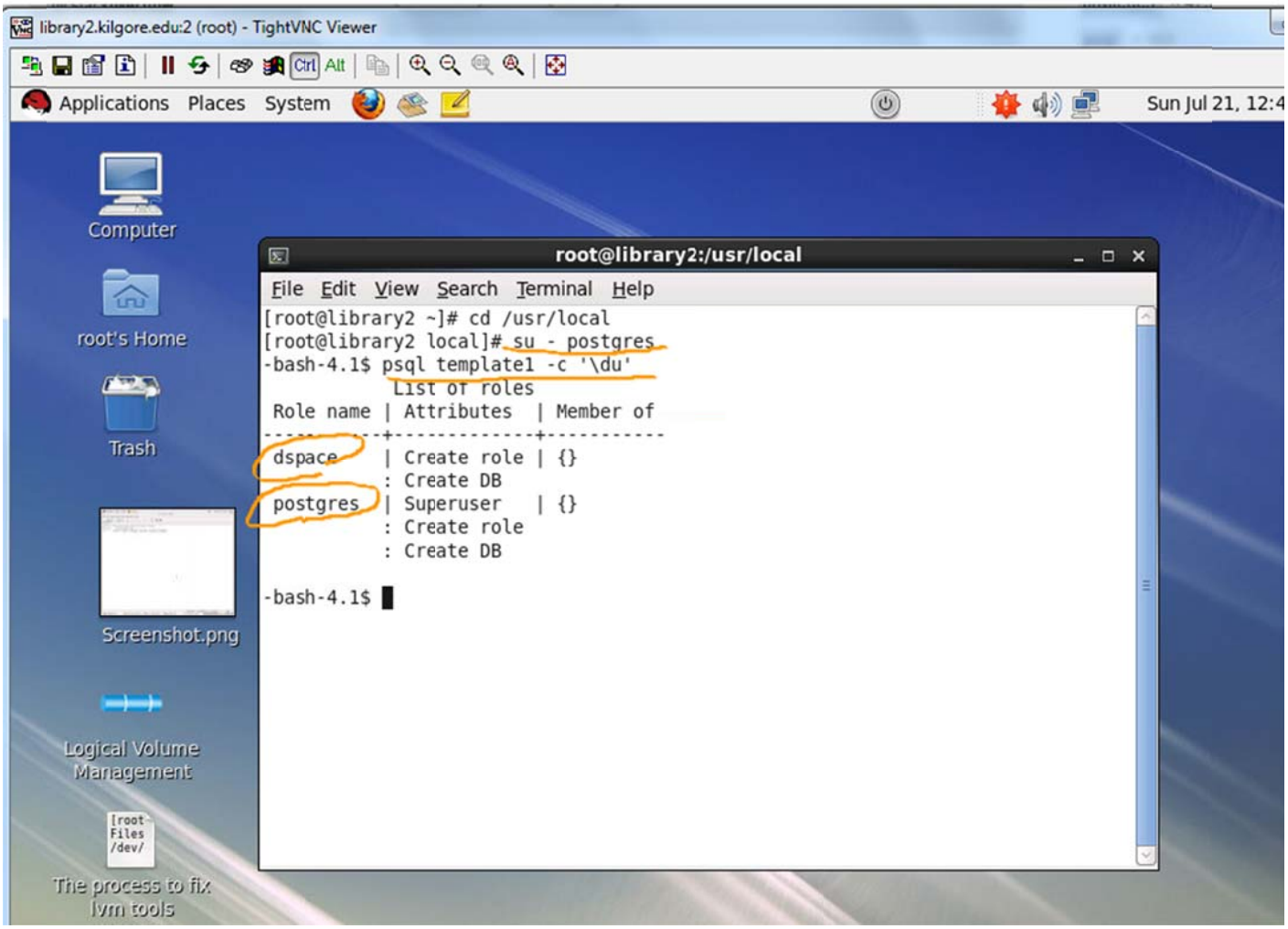
32. Configure other parameters in dspace.cfg file



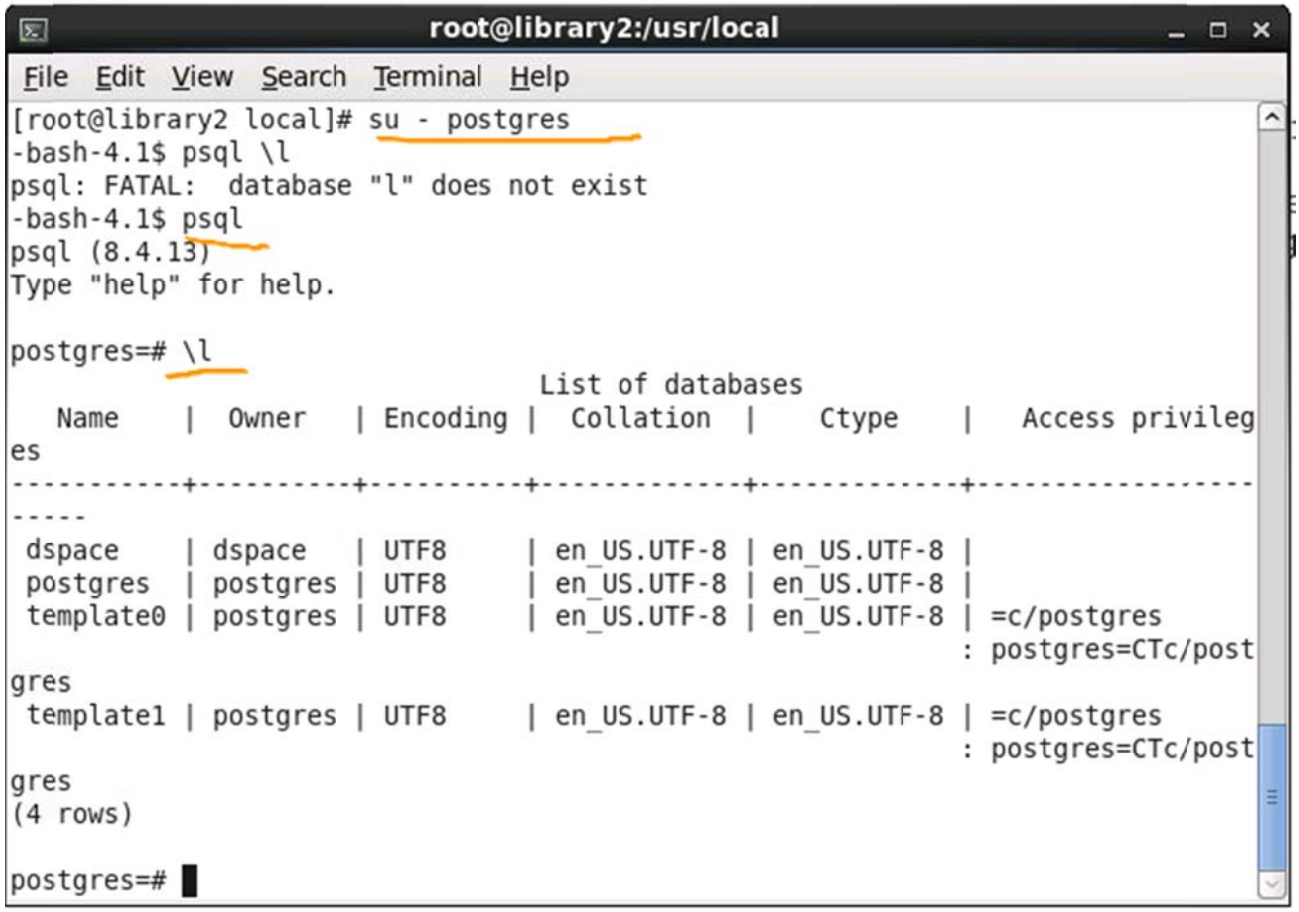
33. Create dspace installation directory



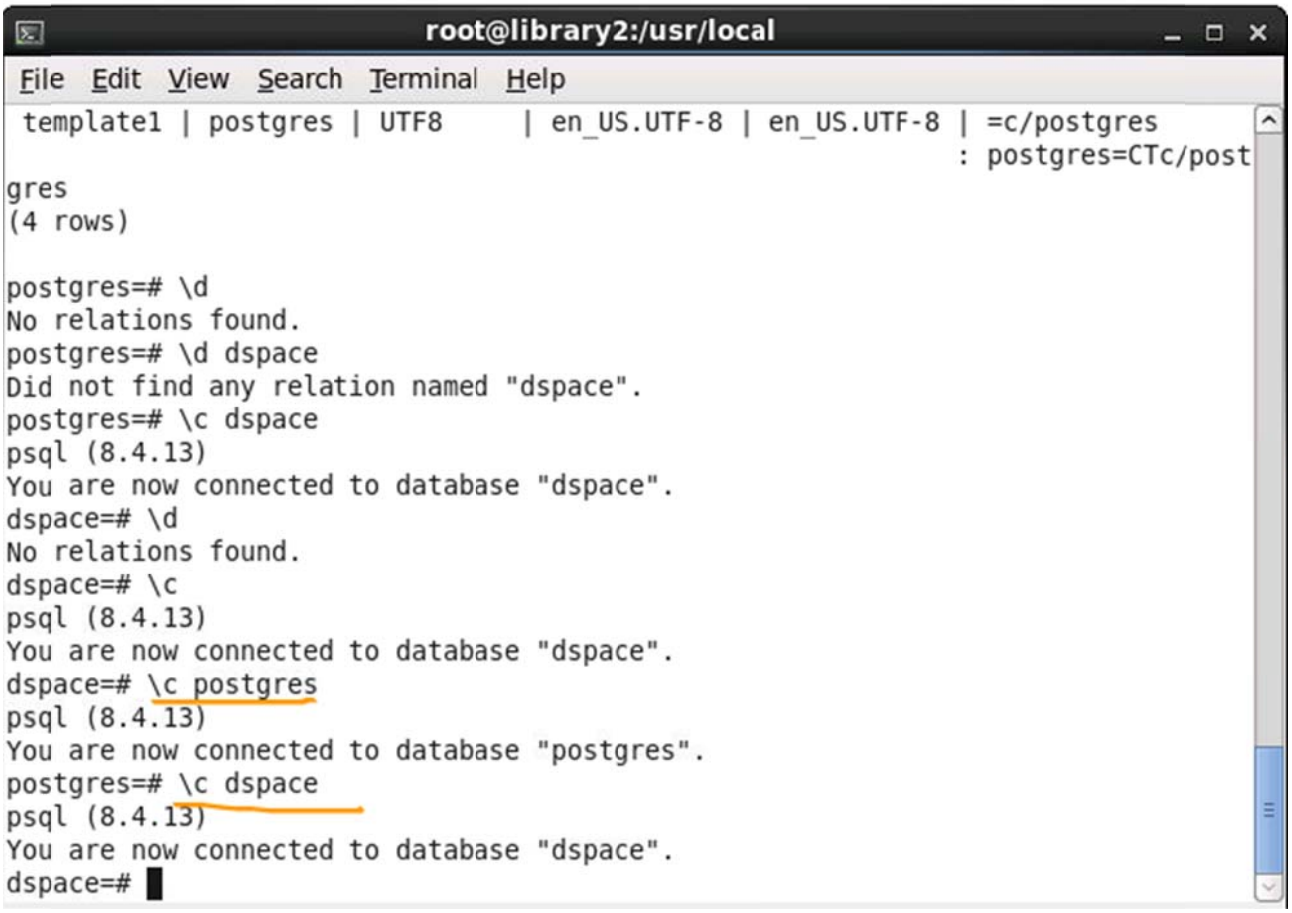
34. List all users in PostgreSQL database



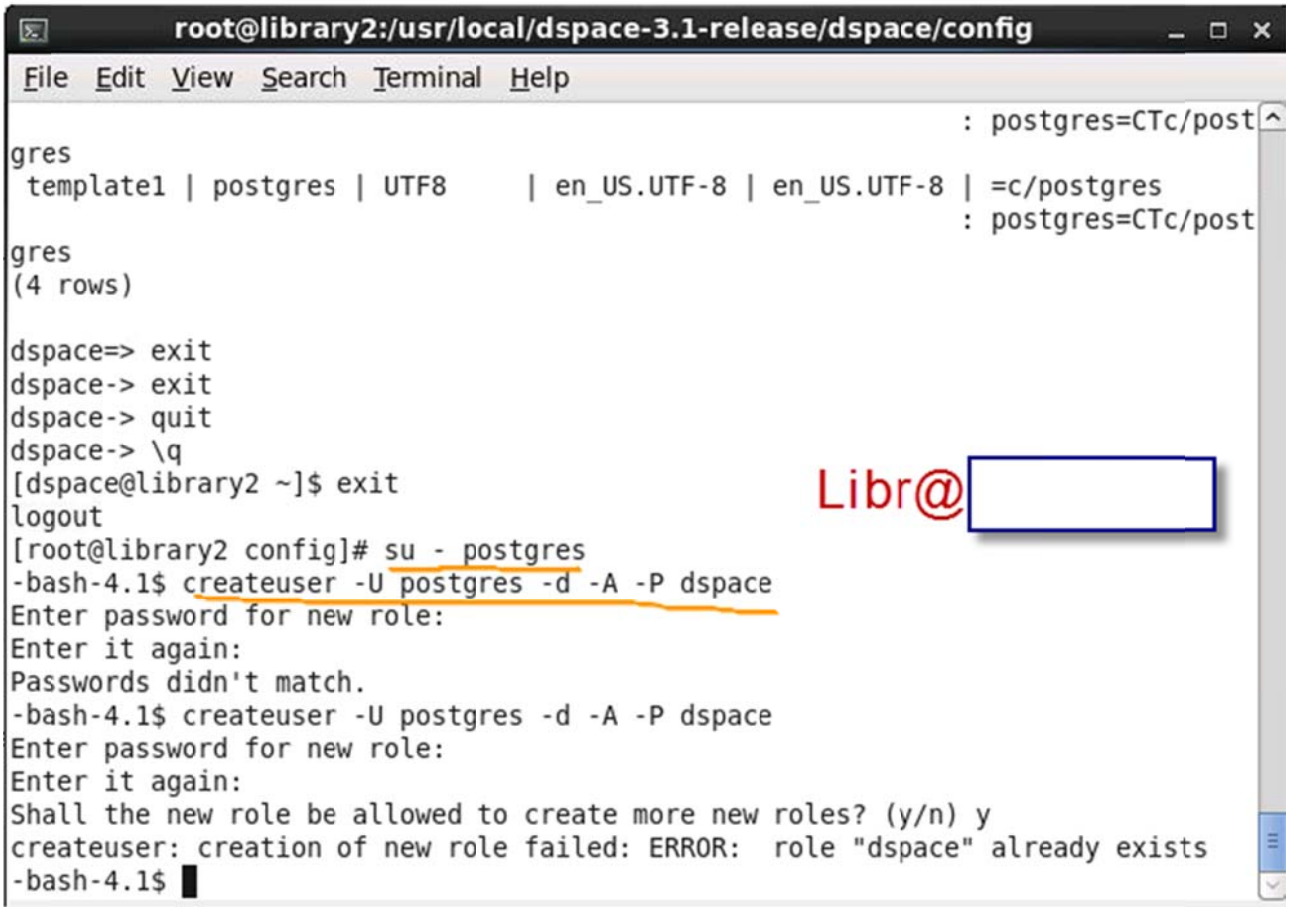
35. List all databases in PostgreSQL



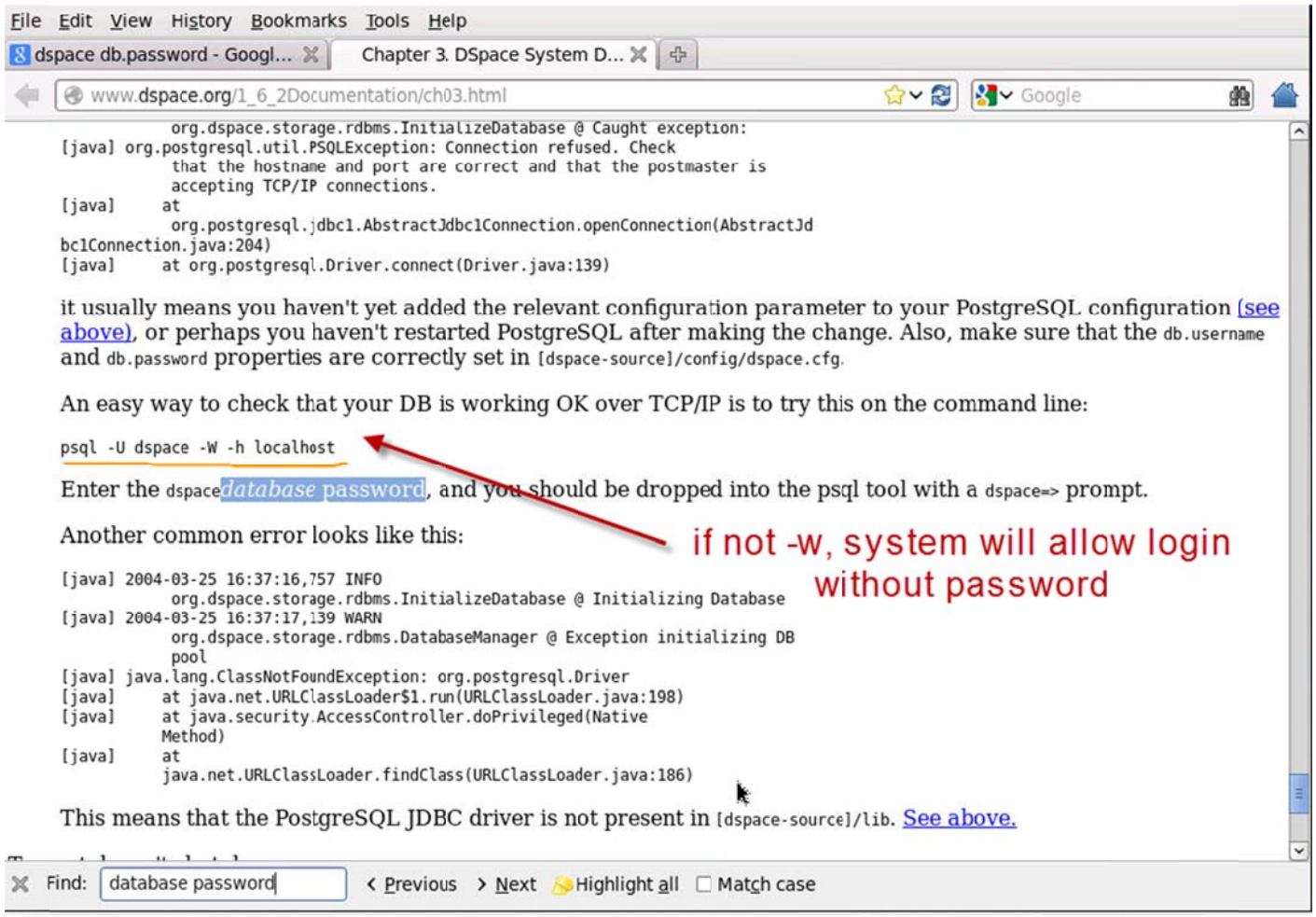
36. How to connect to databases in PostgreSQL



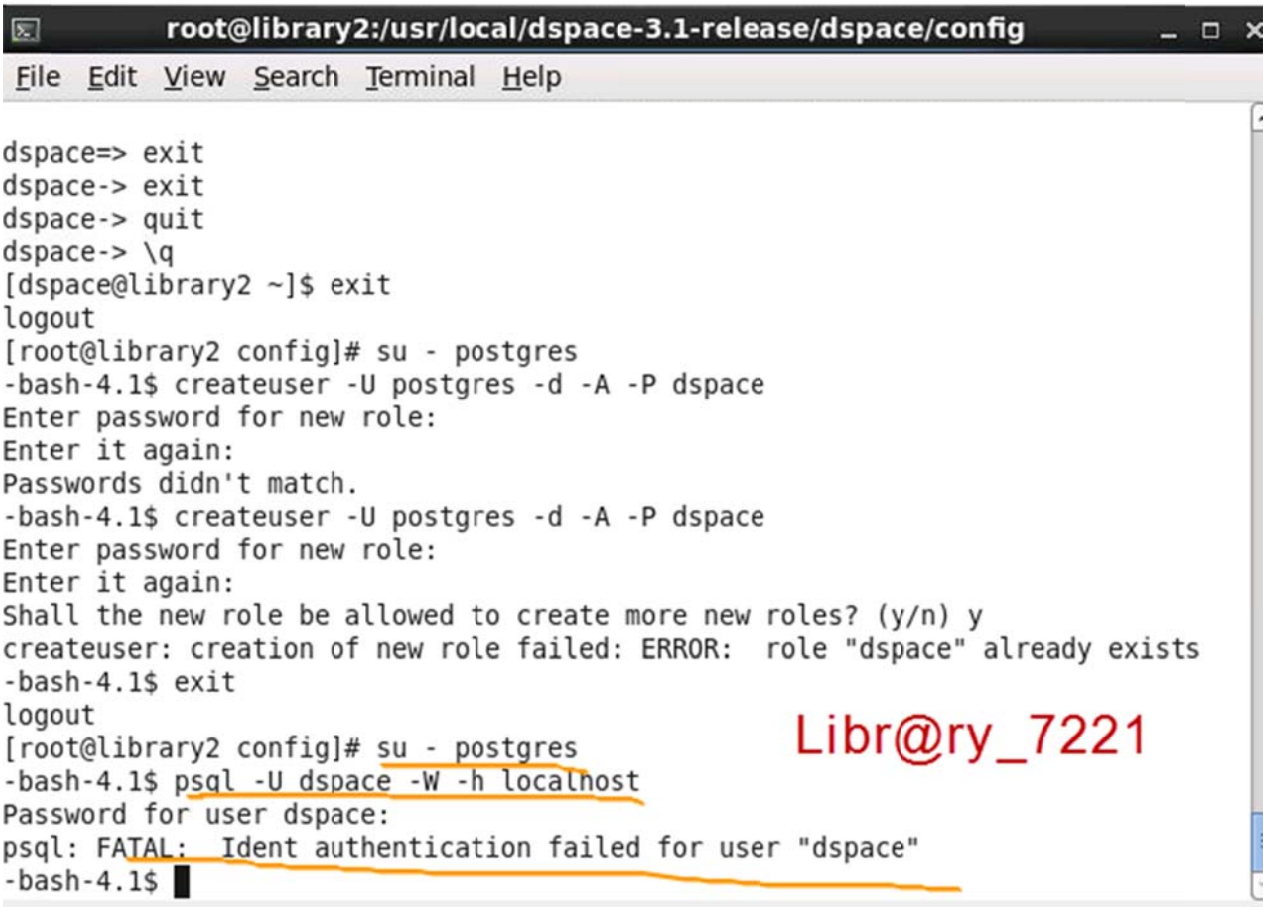
37. Login PostgreSQL database by using user postgres



38. The parameter -w is very important.

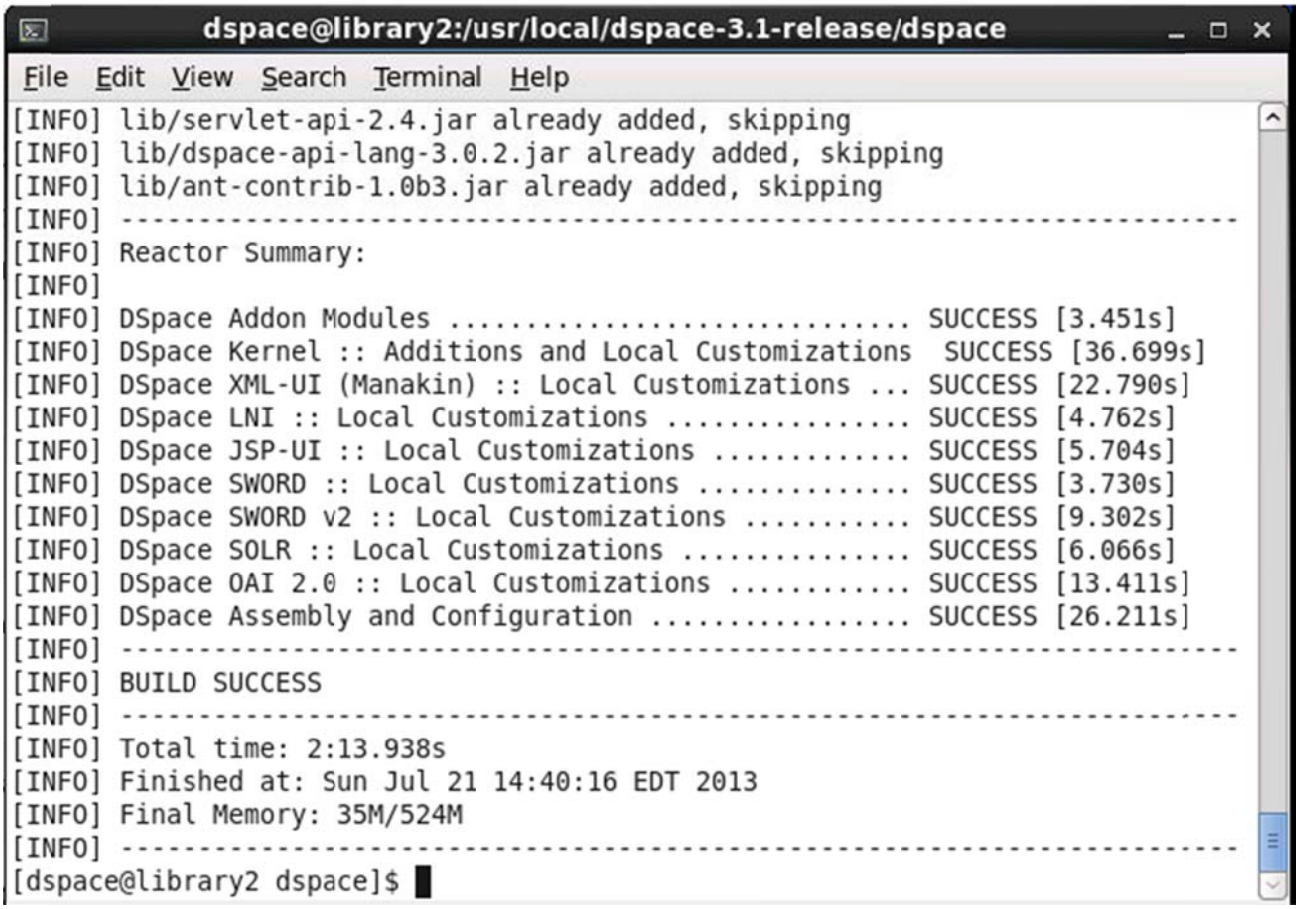


39. The password of user dspace is not correct!

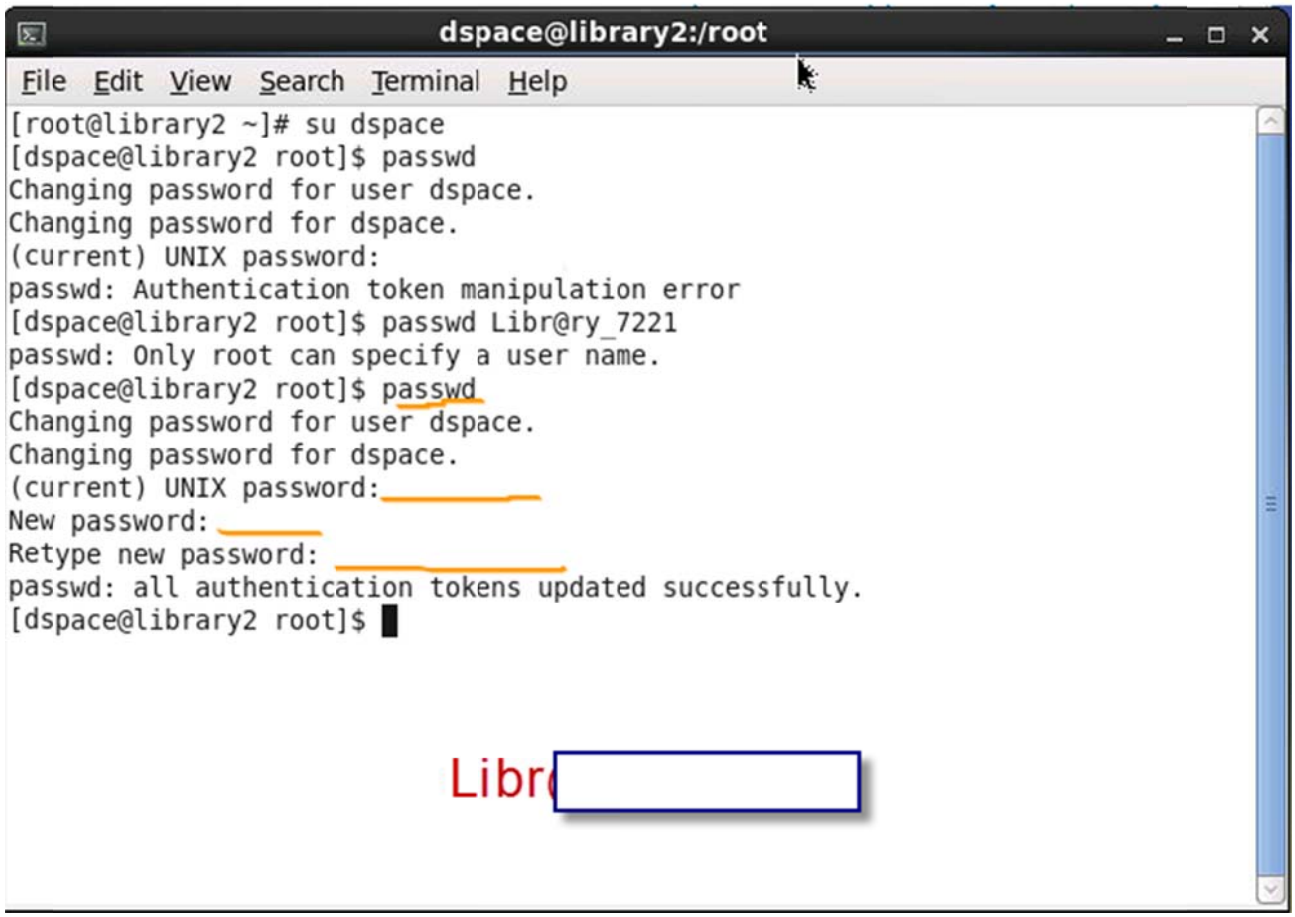


40. Run the command mvn package to build dspace

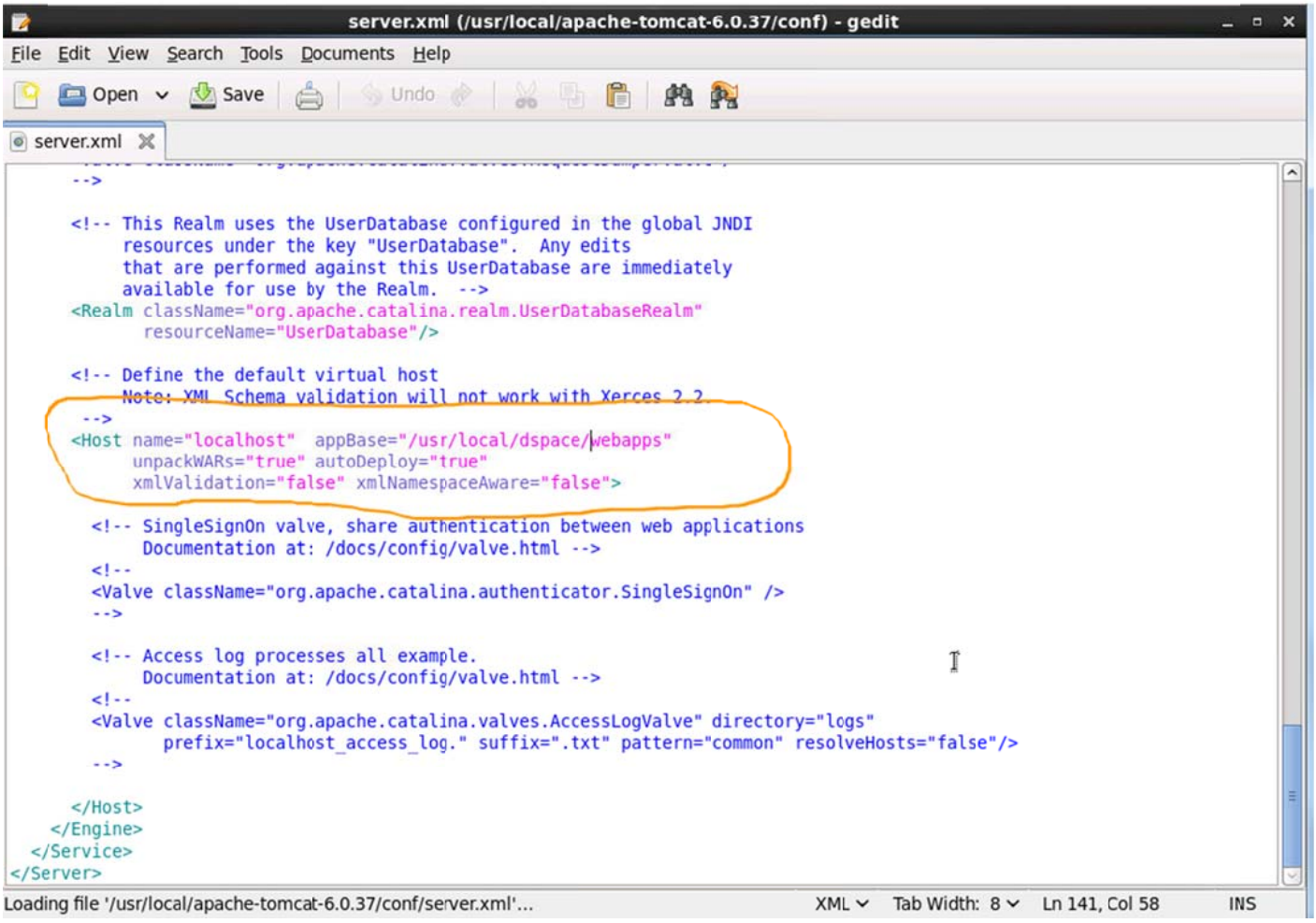
```
su – dspace
cd [dspace-source]/dspace
mvn package
```



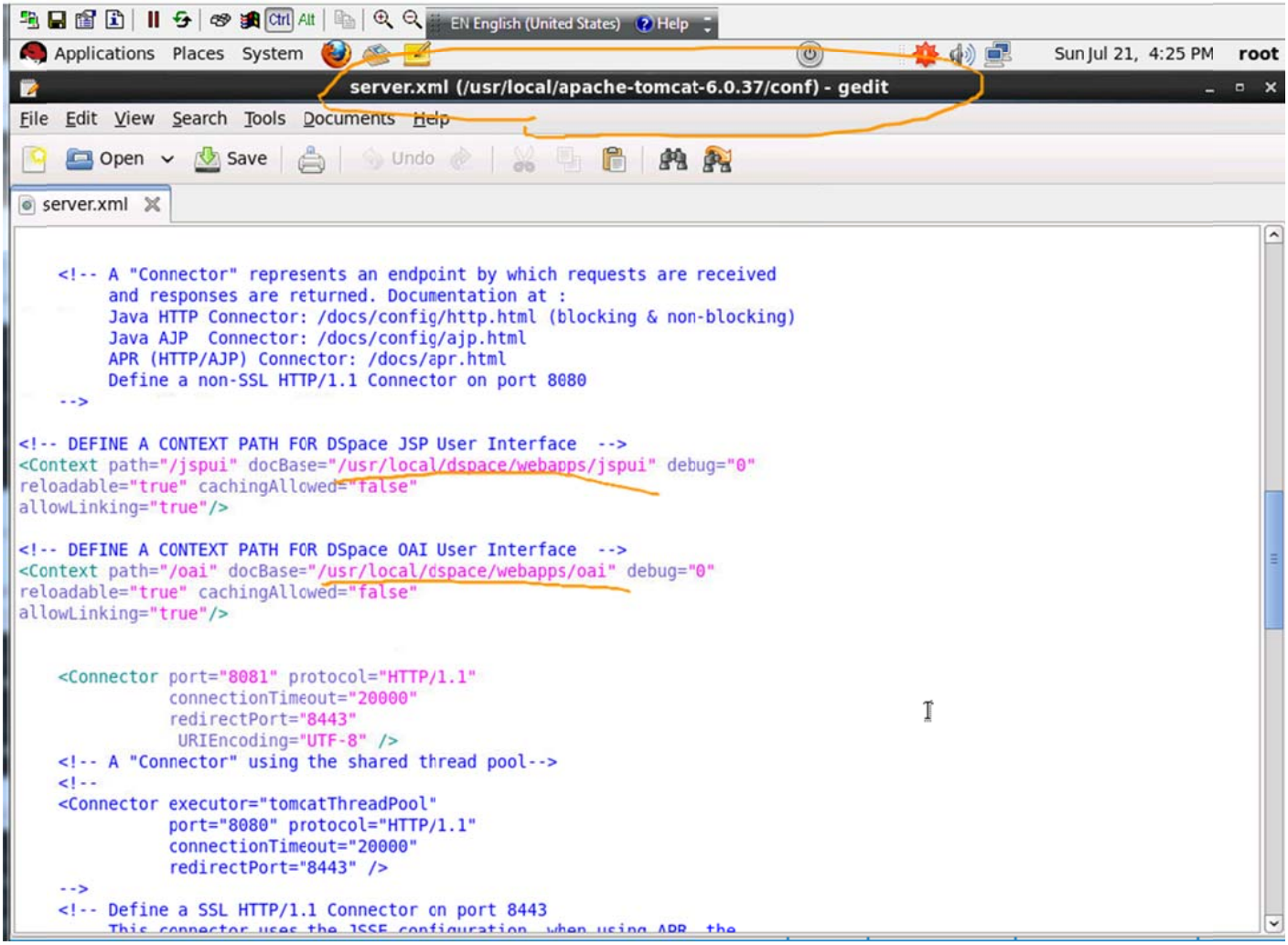
41. Setup dspace user password



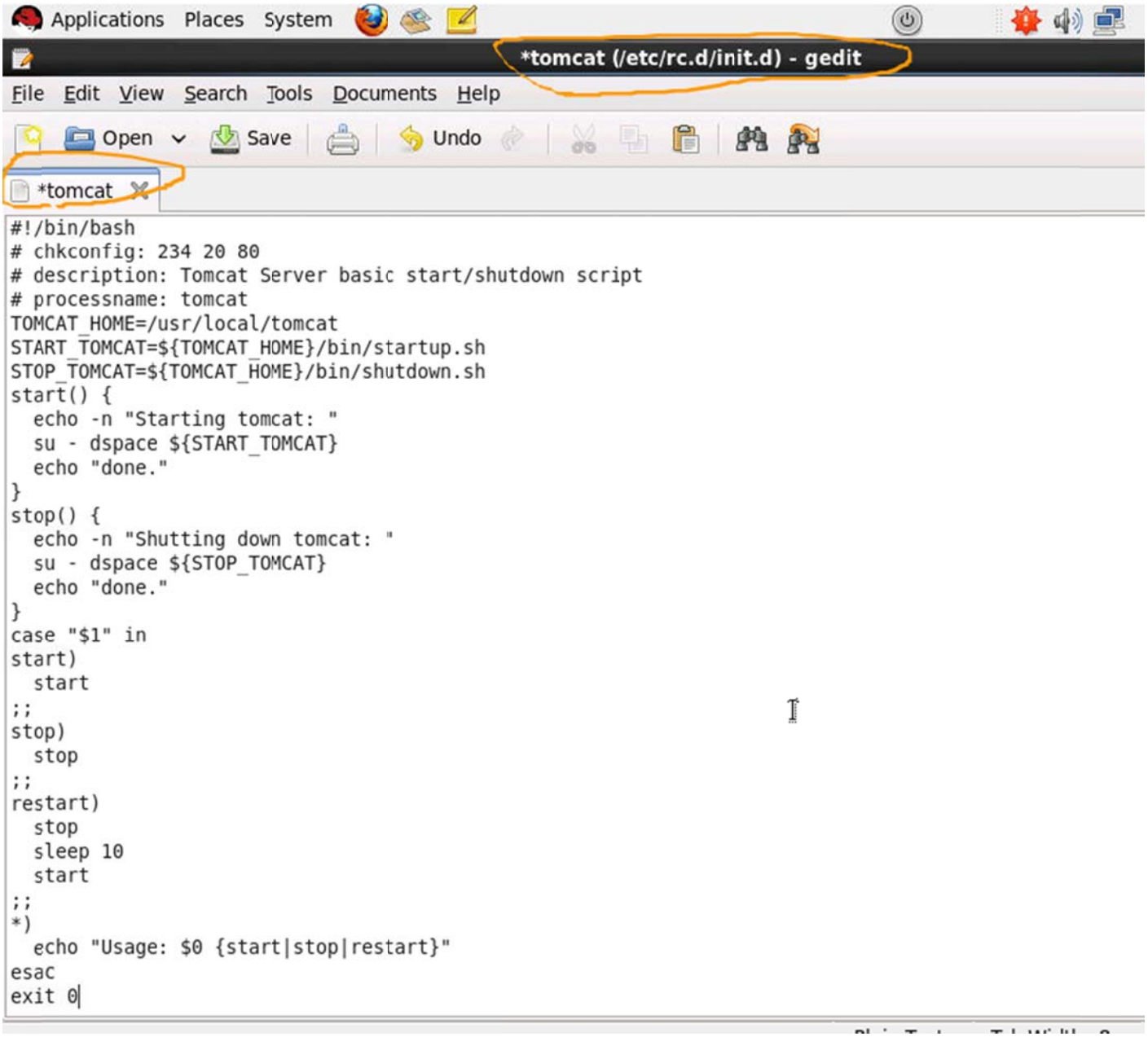
42. Modify server.xml file in tomcat



43. Remember the docbase path

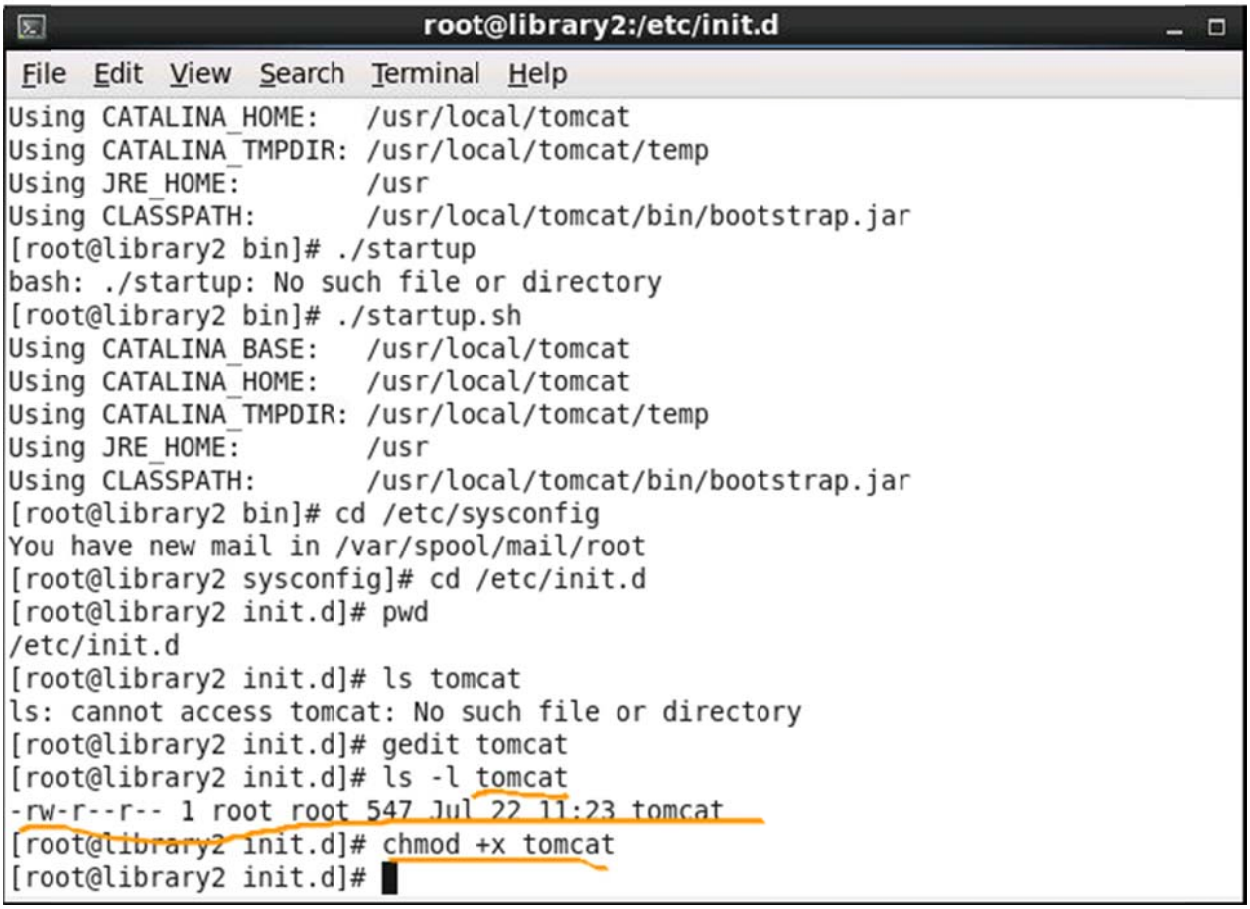


44. Configure Tomcat script in /etc/rc.d/init.d to make it start automatcally when rebooting

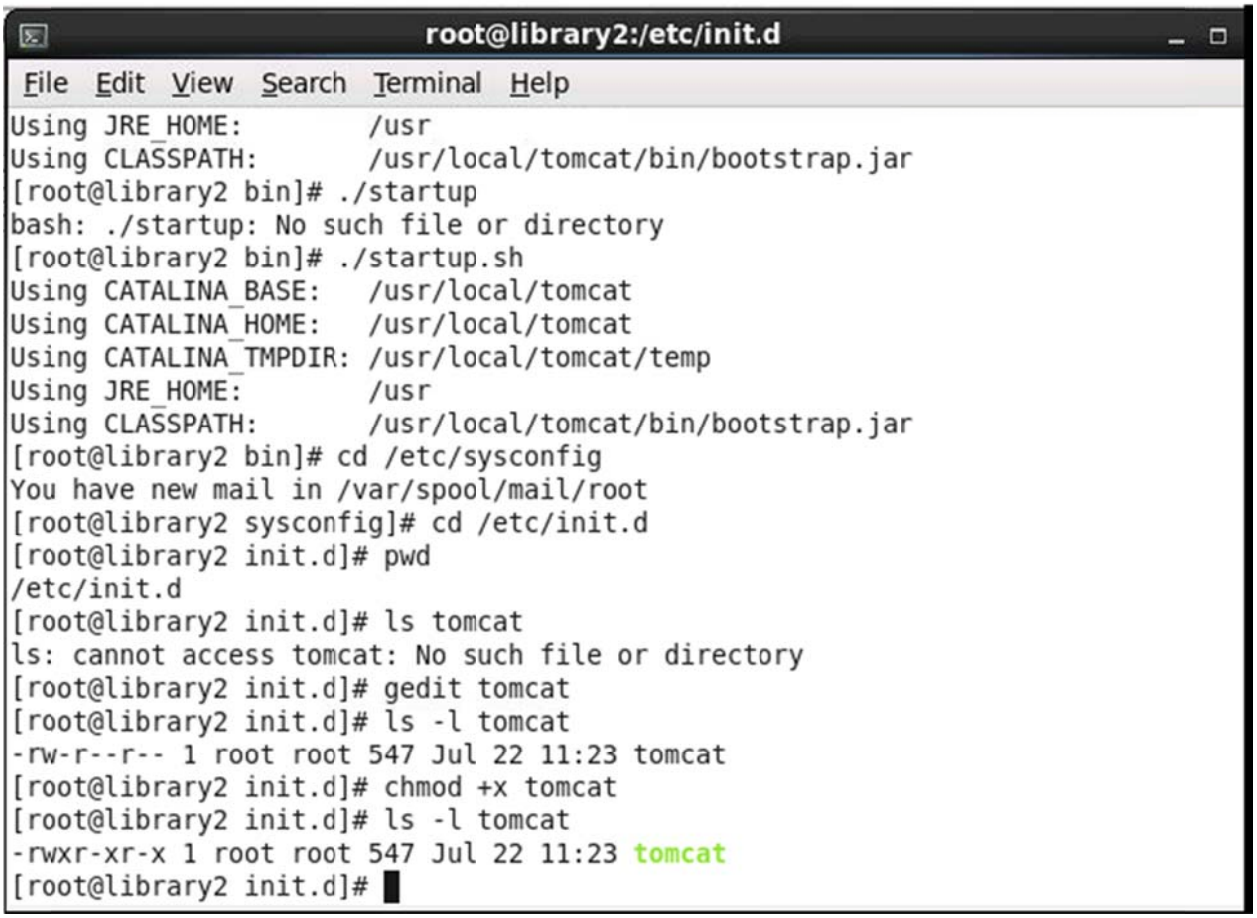



```
#!/bin/bash
# chkconfig: 234 20 80
# description: Tomcat Server basic start/shutdown script
# processname: tomcat
TOMCAT_HOME=/usr/local/tomcat
START_TOMCAT=${TOMCAT_HOME}/bin/startup.sh
STOP_TOMCAT=${TOMCAT_HOME}/bin/shutdown.sh
start() {
  echo -n "Starting tomcat: "
  su - dspace ${START_TOMCAT}
  echo "done."
}
stop() {
  echo -n "Shutting down tomcat: "
  su - dspace ${STOP_TOMCAT}
  echo "done."
}
case "$1" in
start)
  start
;;
stop)
  stop
;;
restart)
  stop
  sleep 10
  start
;;
*)
  echo "Usage: $0 {start|stop|restart}"
esac
exit 0
```

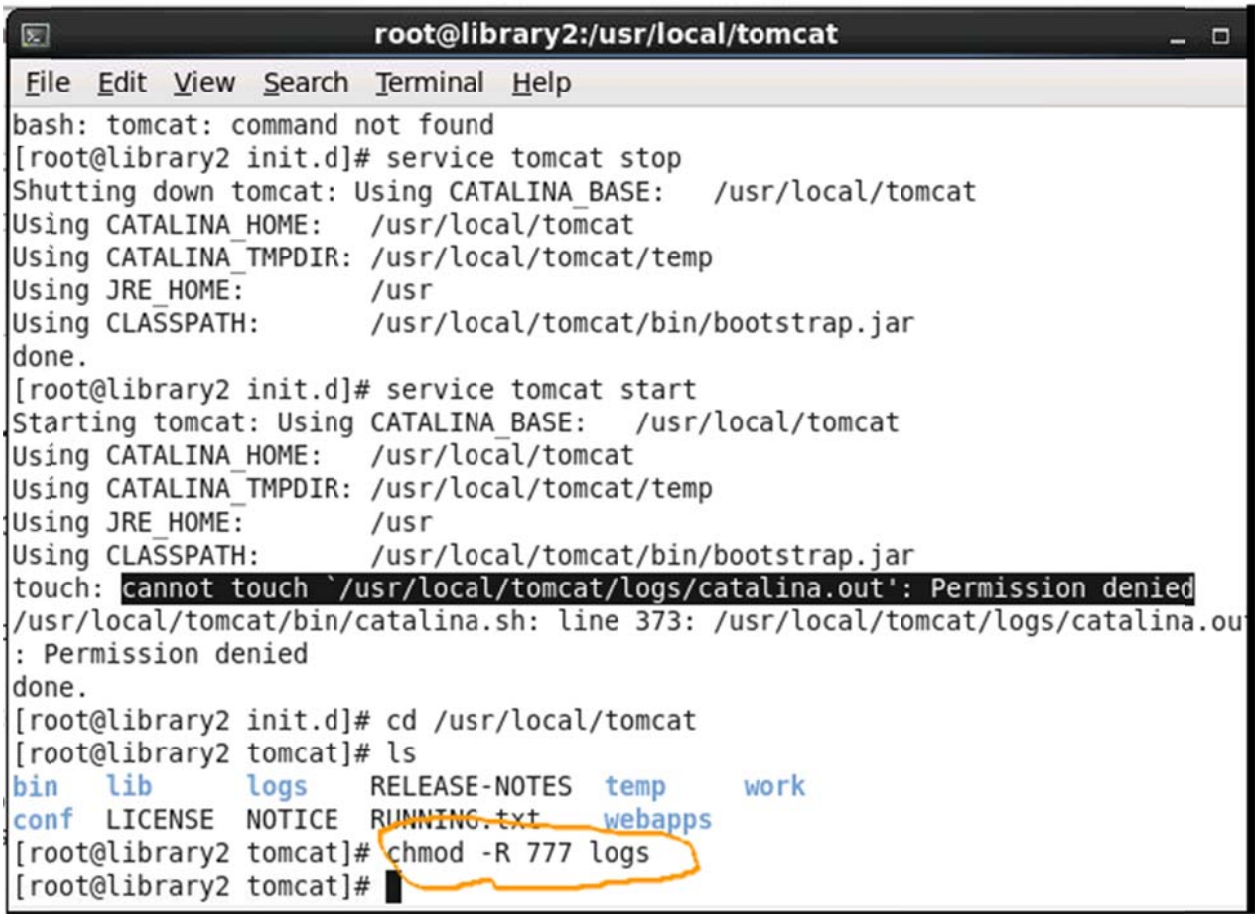
45. Make tomcat script executable



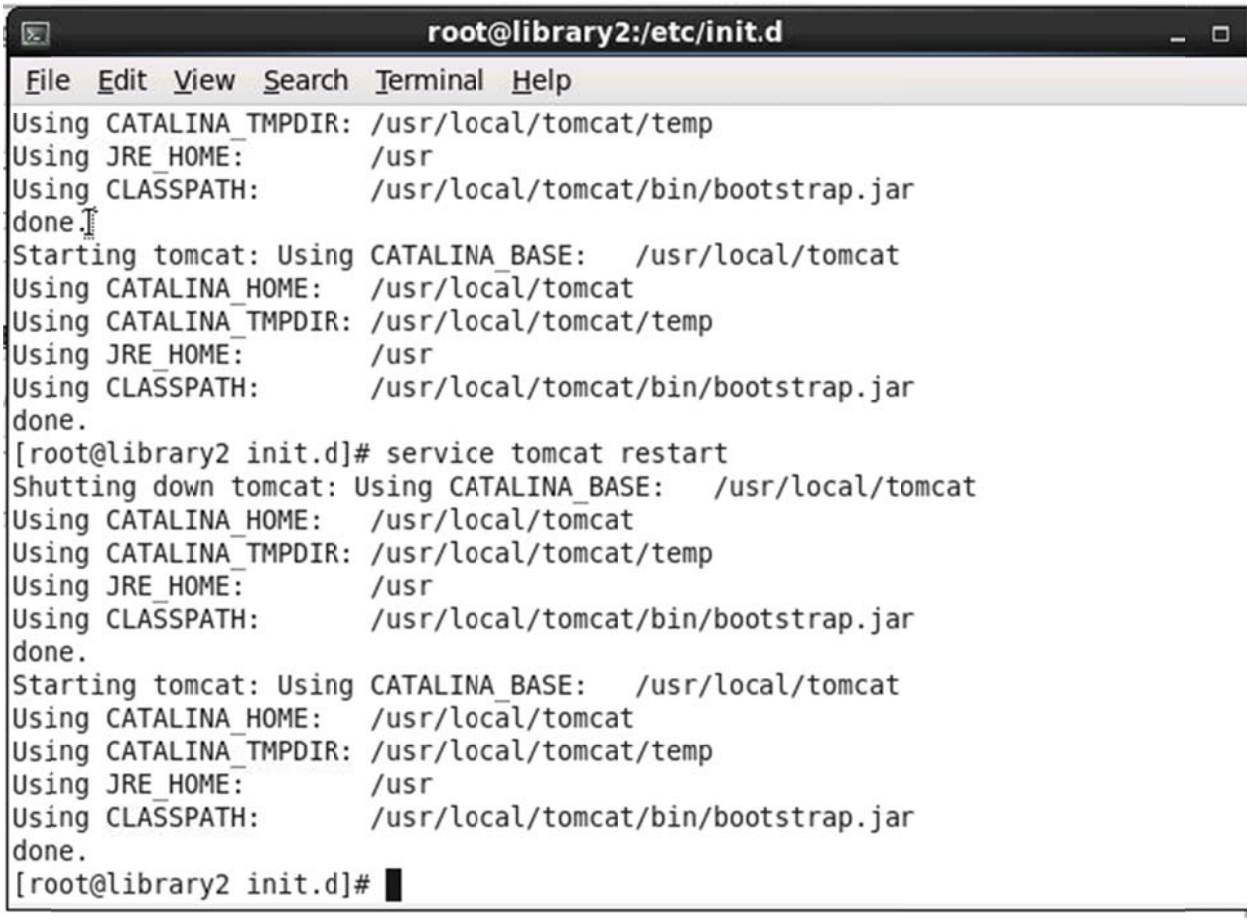
46. After running chmod command, tomcat is executable.



47. When starting tomcat, the problem of cannot touch catalina.out comes out. I need to solve this problem. By using Chmod, /usr/local/tomcat.logs folder is authorized to becoming writable by tomcat program.



If times gap between two "service tomcat restar" is too short, there will be some problem happened!



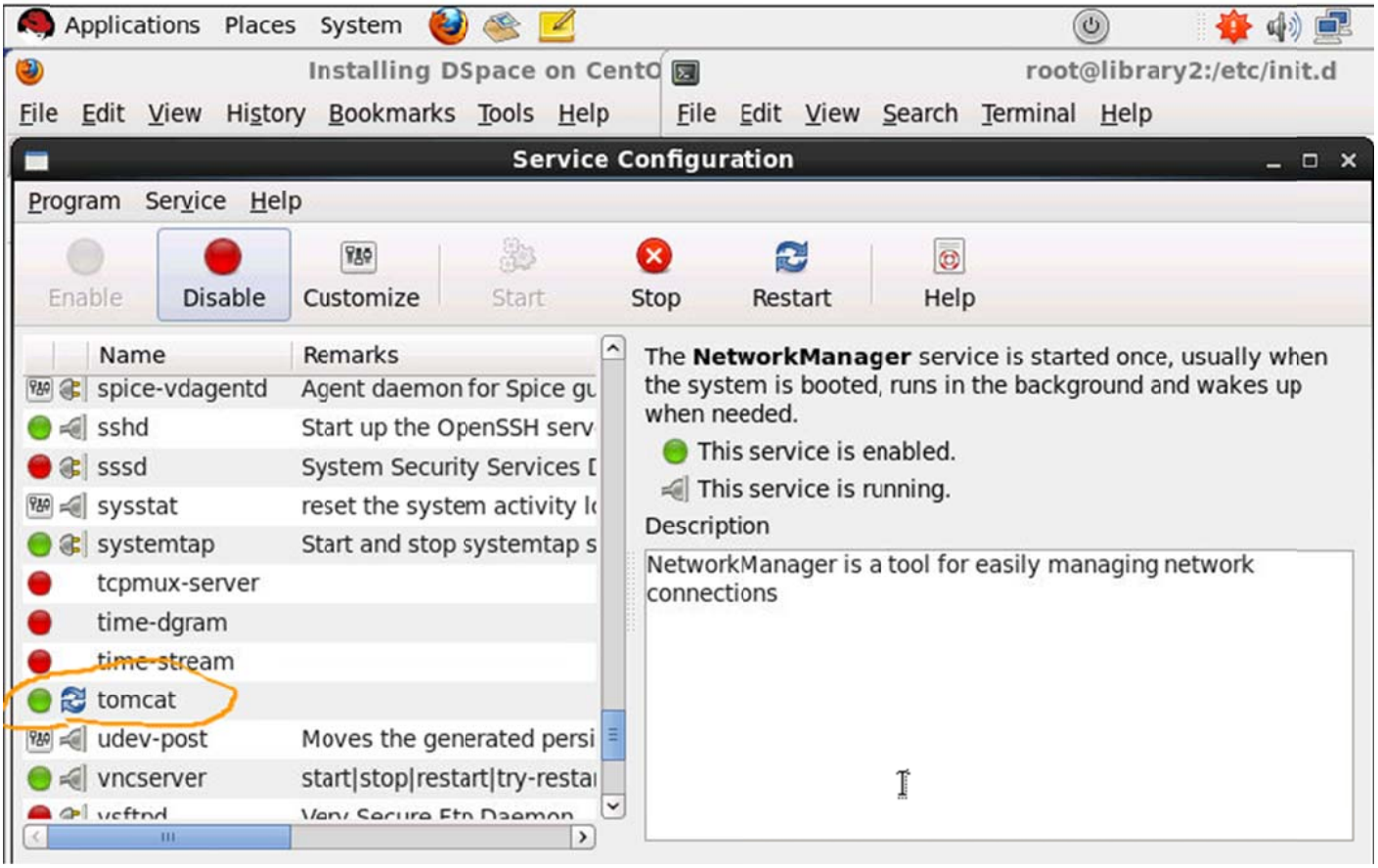
49. Add the tomcat script to chkconfig and set it to run on appropriate system runlevel

```
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
done.
[root@library2 init.d]# service tomcat restart
Shutting down tomcat: Using CATALINA_BASE:   /usr/local/tomcat
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
done.
Starting tomcat: Using CATALINA_BASE:   /usr/local/tomcat
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
done.
[root@library2 init.d]# chkconfig --add tomcat
[root@library2 init.d]# chkconfig --level 234 tomcat on
[root@library2 init.d]#
```

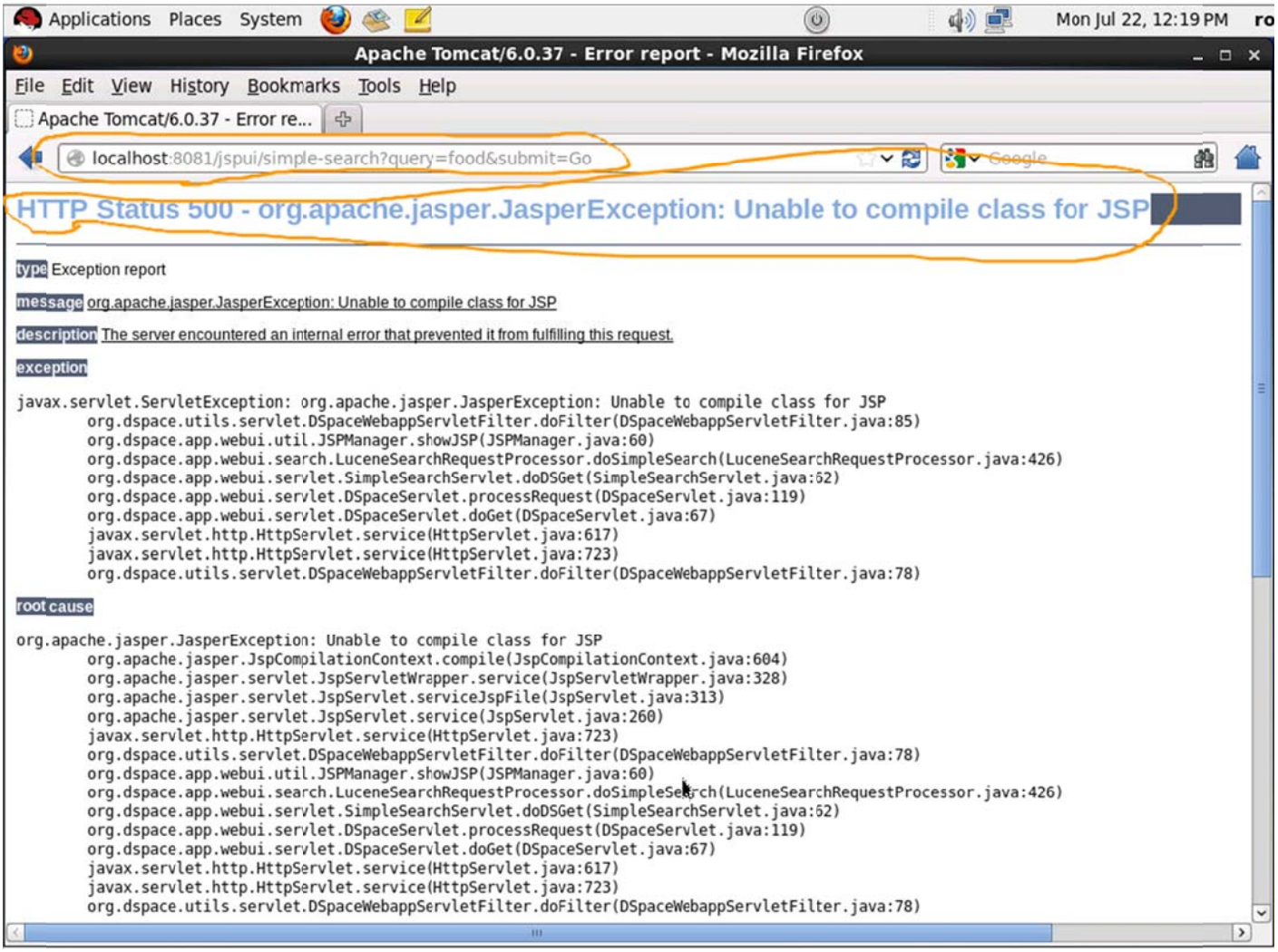
50. Check the runlevel of tomcat script

```
done.
[root@library2 init.d]# service tomcat restart
Shutting down tomcat: Using CATALINA_BASE:   /usr/local/tomcat
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
done.
Starting tomcat: Using CATALINA_BASE:   /usr/local/tomcat
Using CATALINA_HOME:   /usr/local/tomcat
Using CATALINA_TMPDIR: /usr/local/tomcat/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /usr/local/tomcat/bin/bootstrap.jar
done.
[root@library2 init.d]# chkconfig --add tomcat
[root@library2 init.d]# chkconfig --level 234 tomcat on
[root@library2 init.d]# chkconfig --list tomcat
tomcat          0:off   1:off   2:on    3:on    4:on    5:off   6:off
[root@library2 init.d]#
```

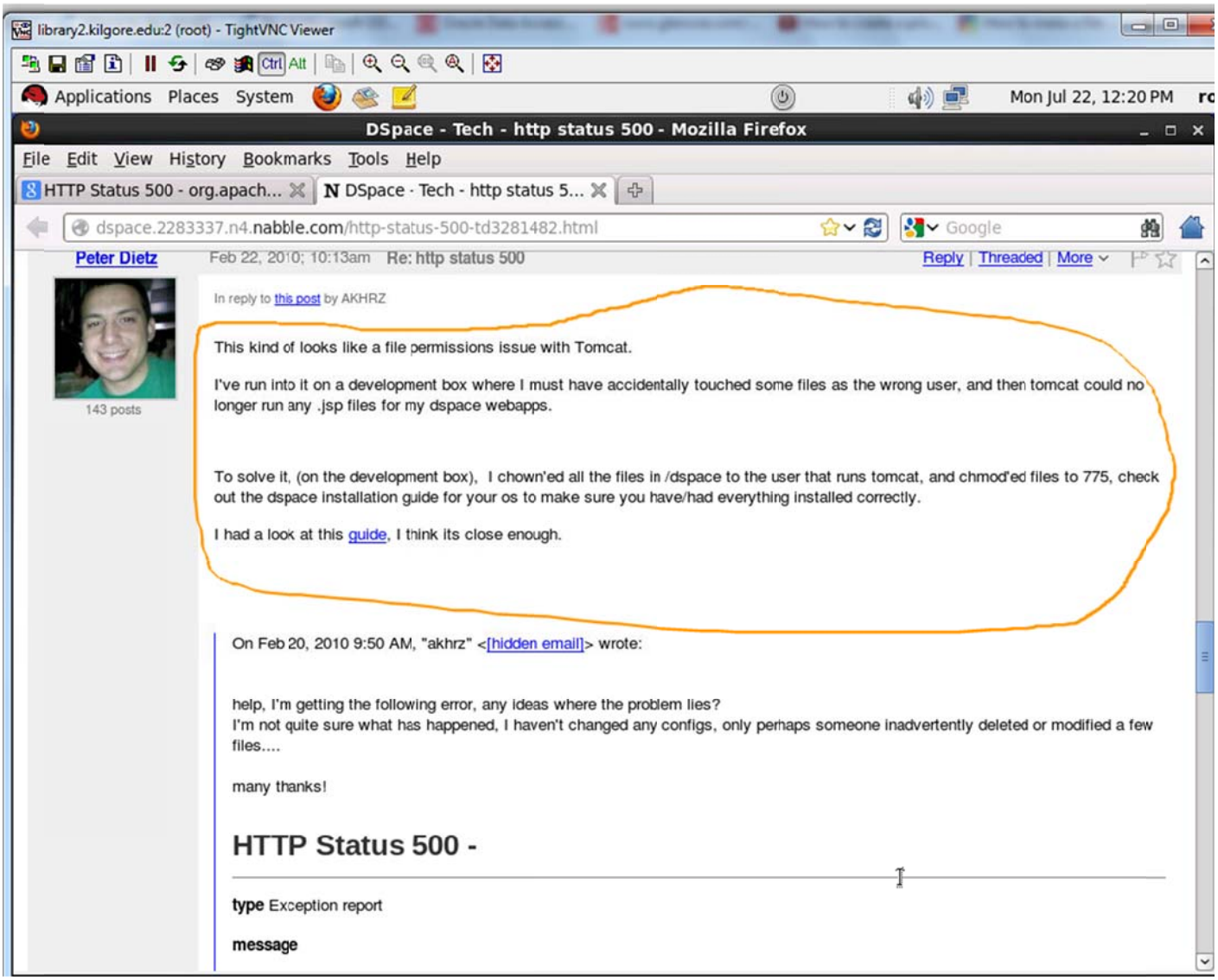
51. Tomcat is in service now



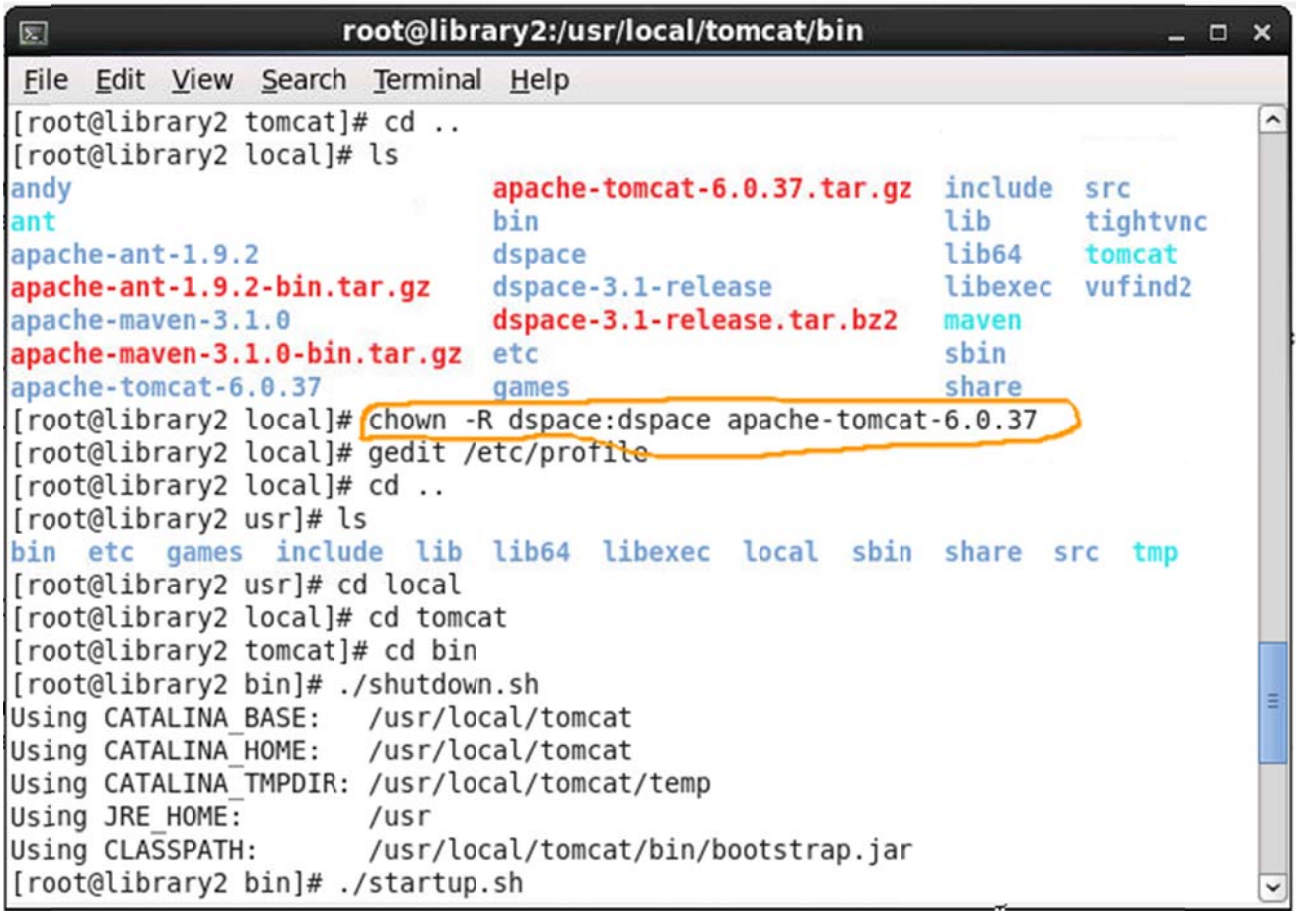
52. However, after installation, dspace discovery interface has http status 500 problem. It is not a big deal. Let us solve it.



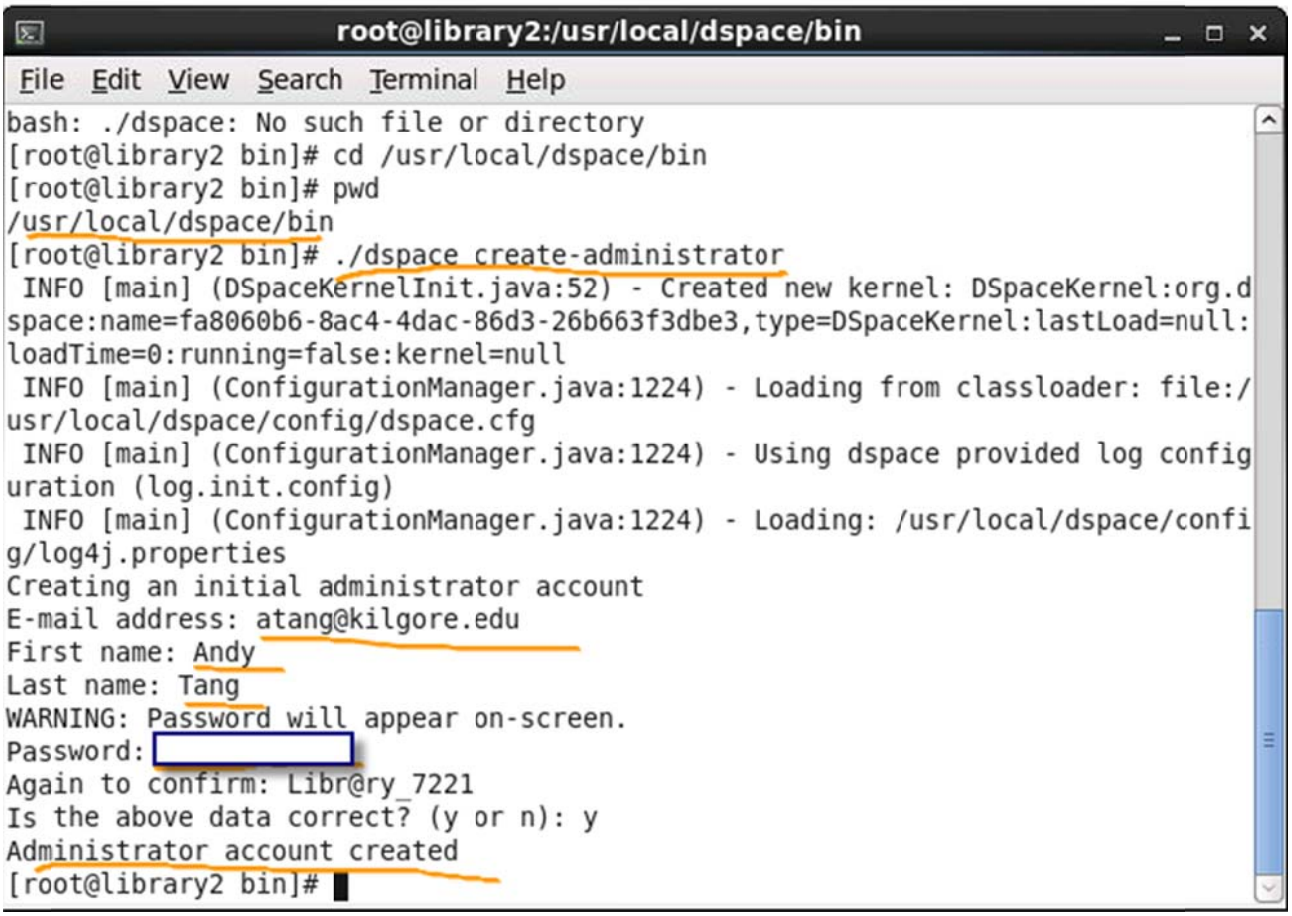
53. In Dspace listserver, Peter Dietz gave a solution on this situation which run chomd and gave 775 authority to all files under the dspace folder



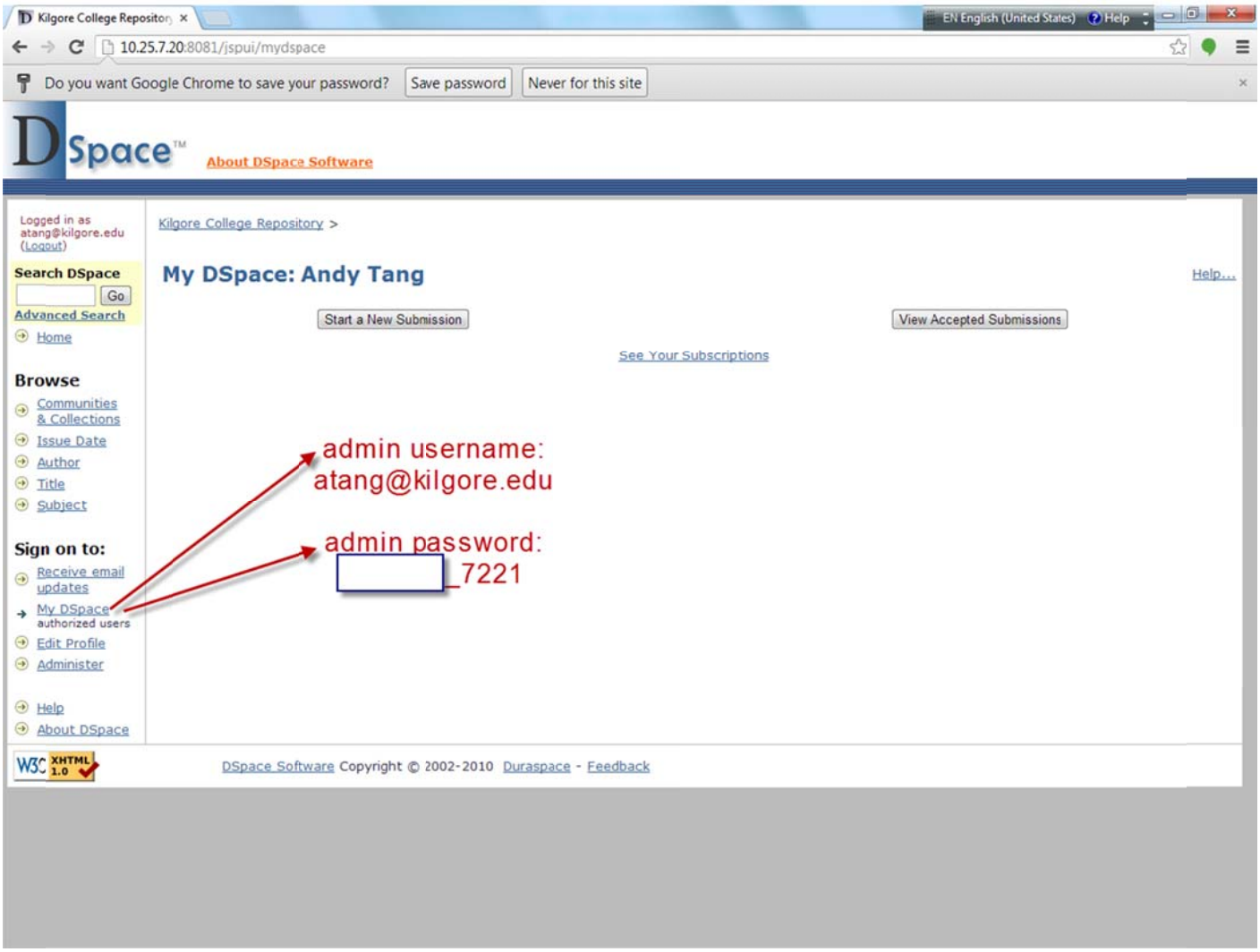
54. I redo the chown command for apache-tomcat-6.037 folder



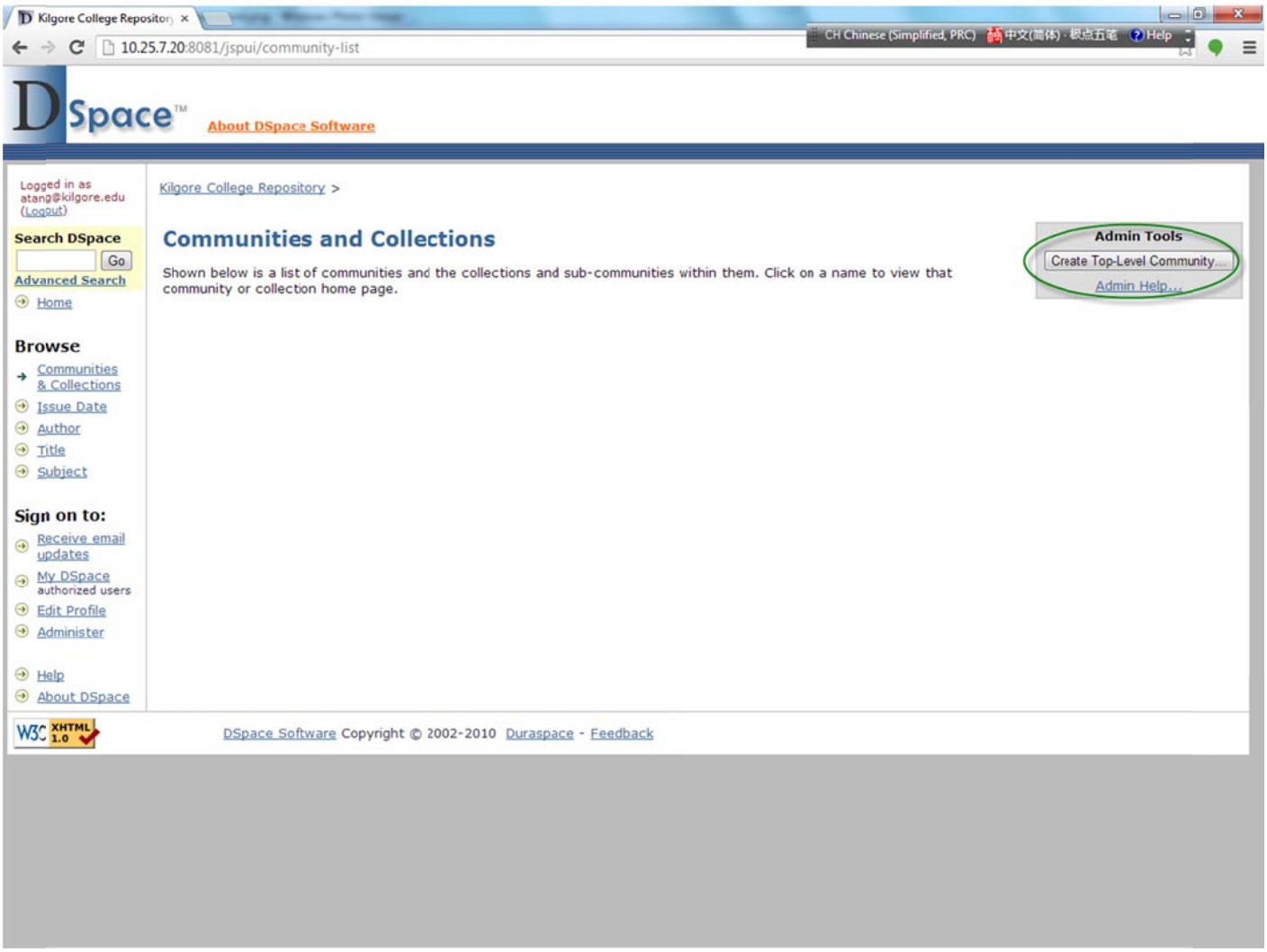
55. The administrator and password in dspace are created



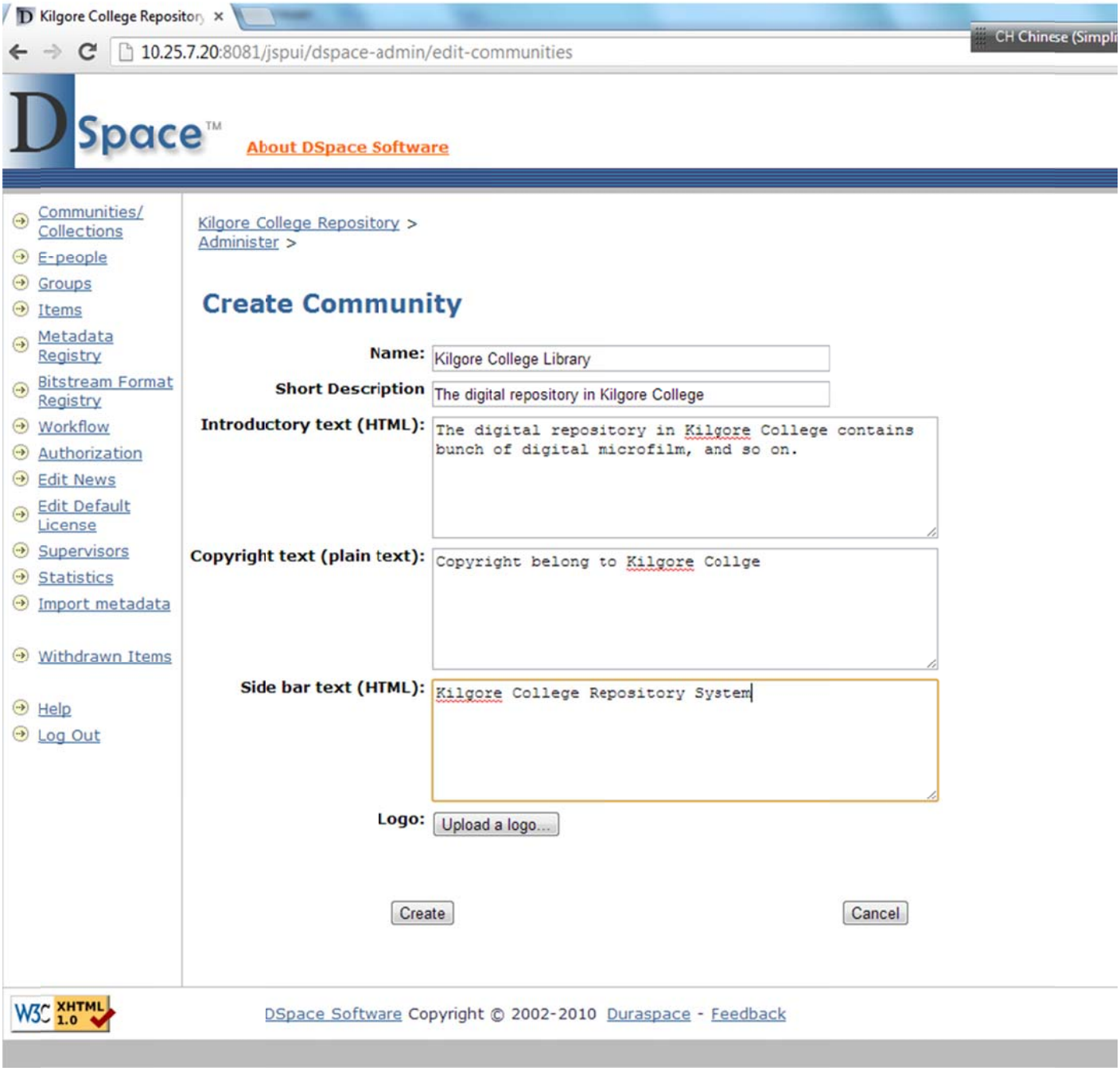
56. Login Dspace by using username and password we created



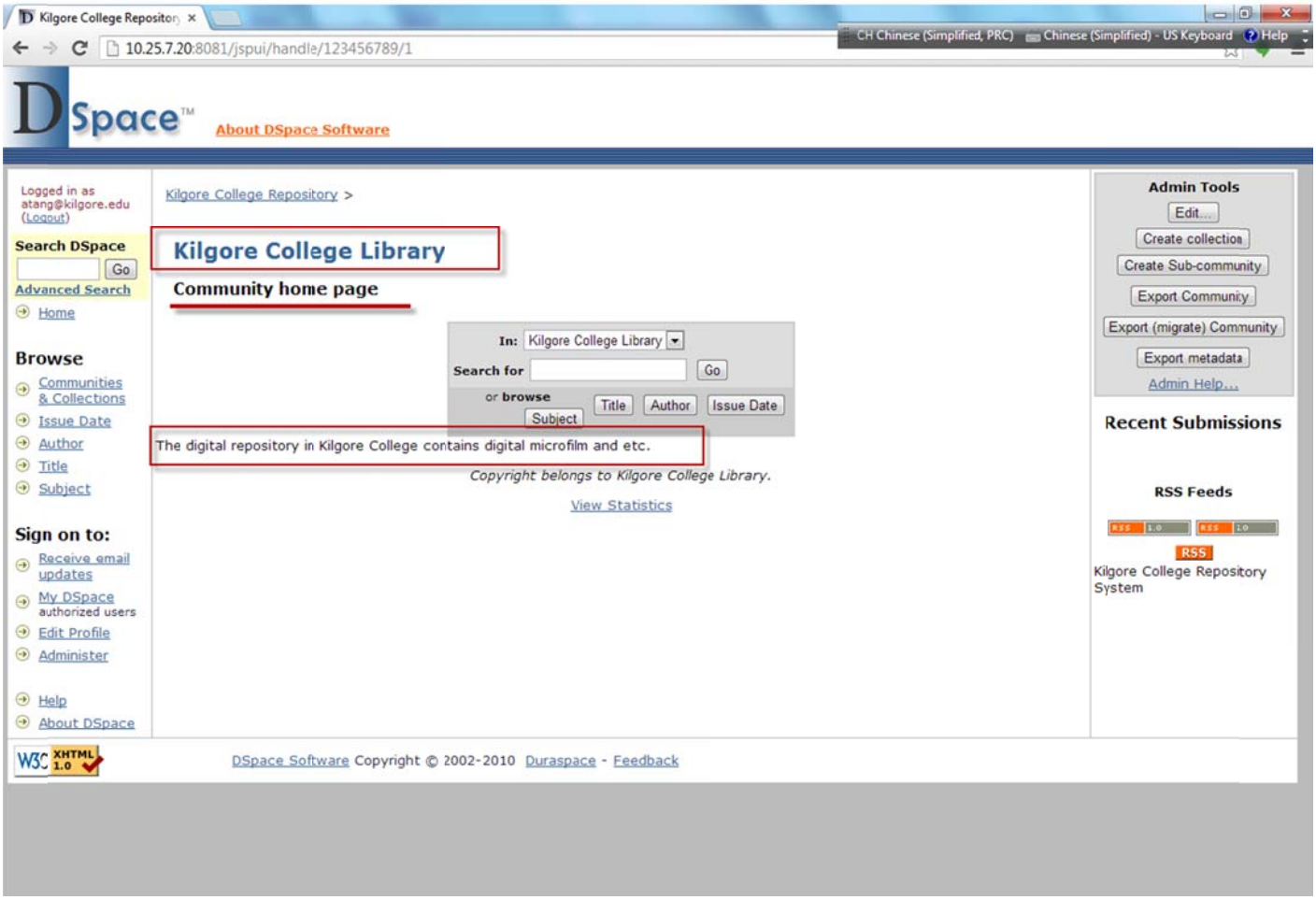
57. Create community button



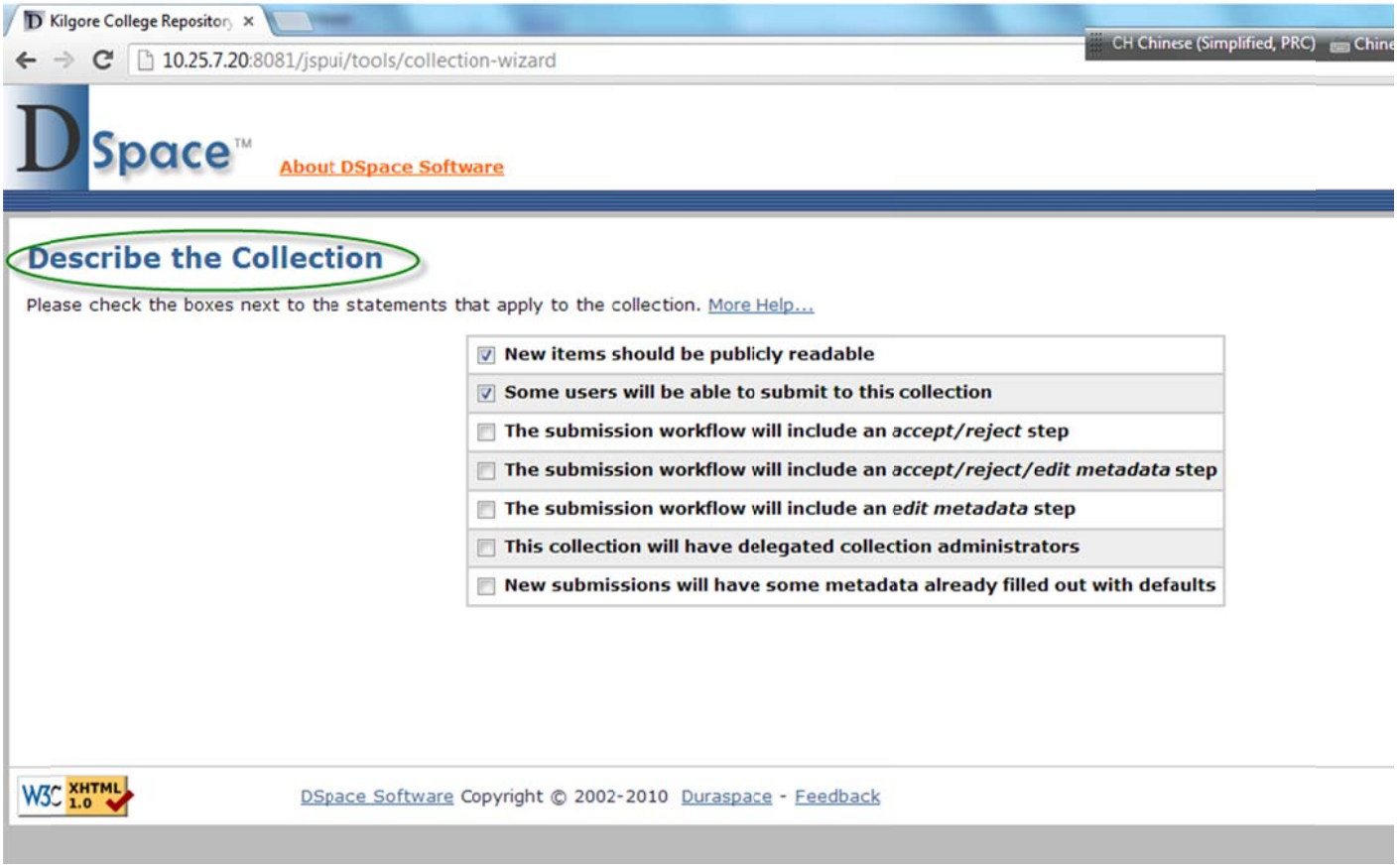
58. Create community step by step



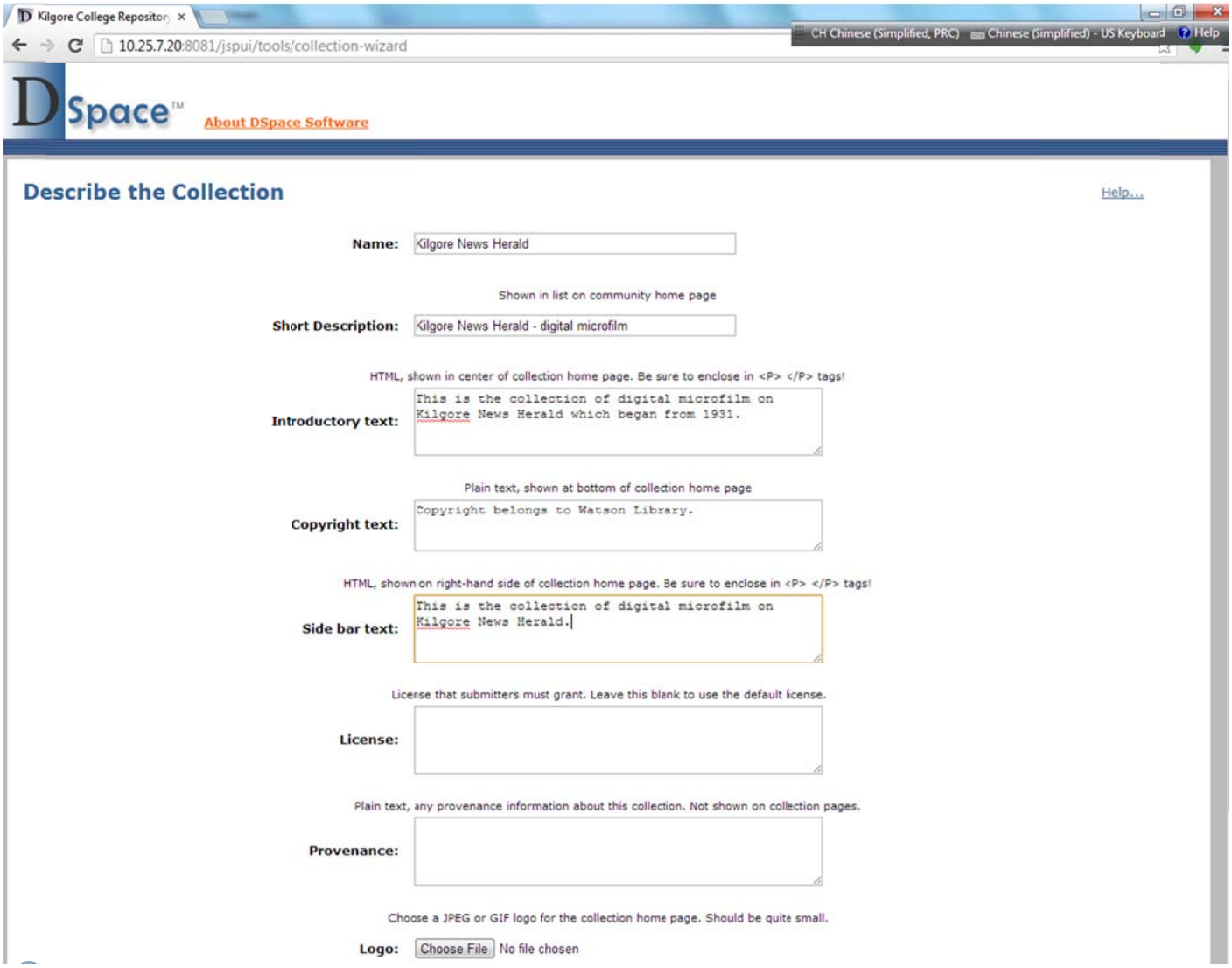
59. Community home page is created



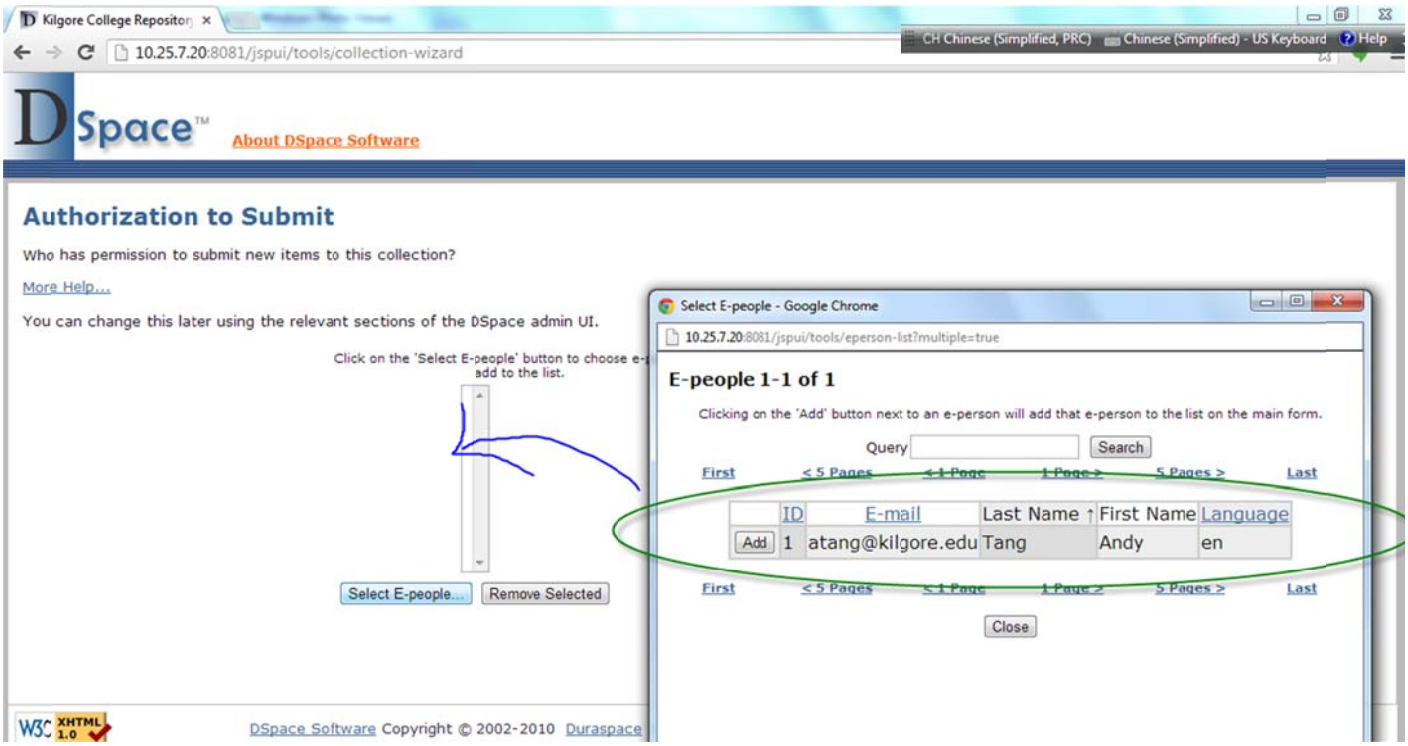
60. Describe the collection



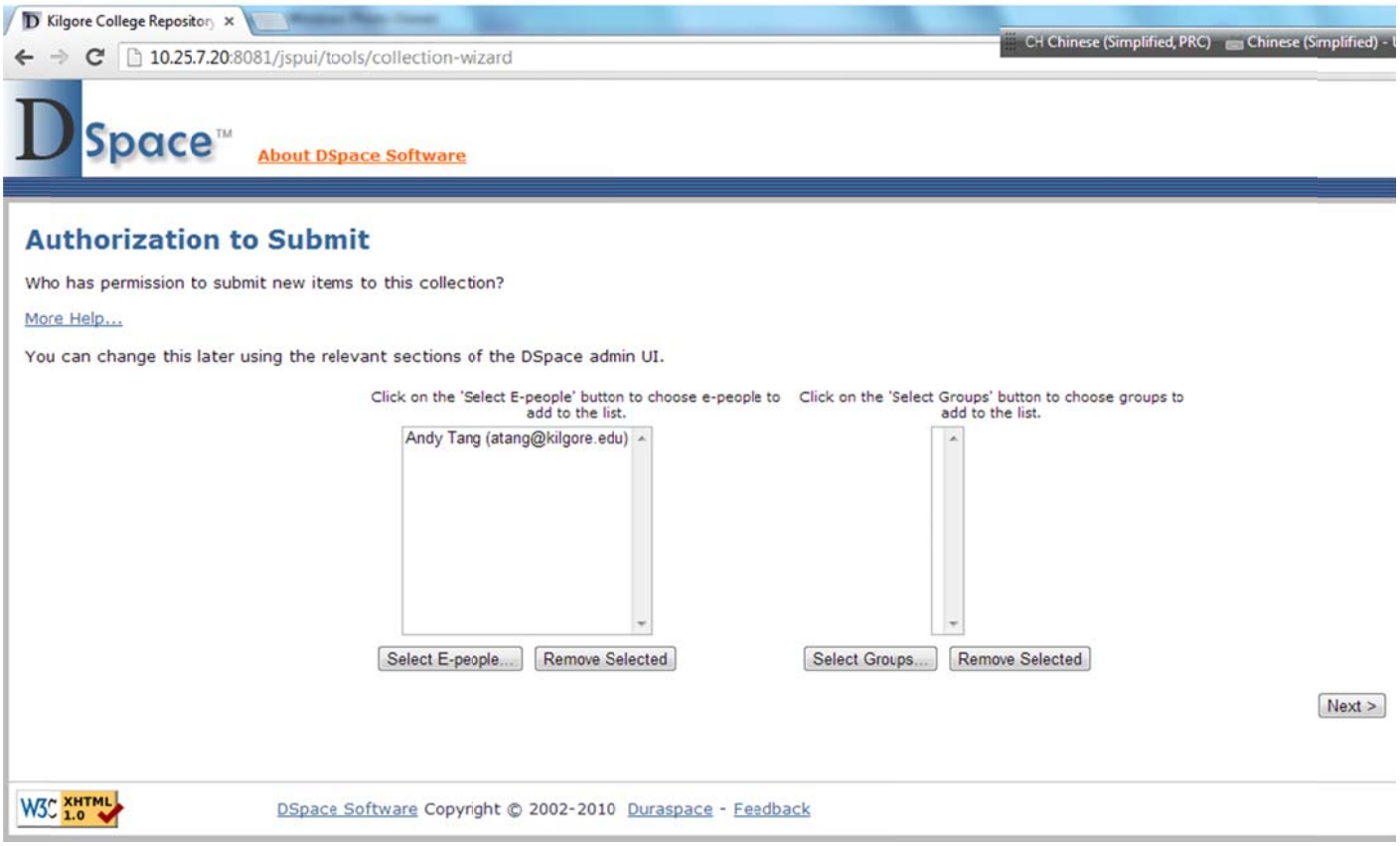
61. Describe the collection



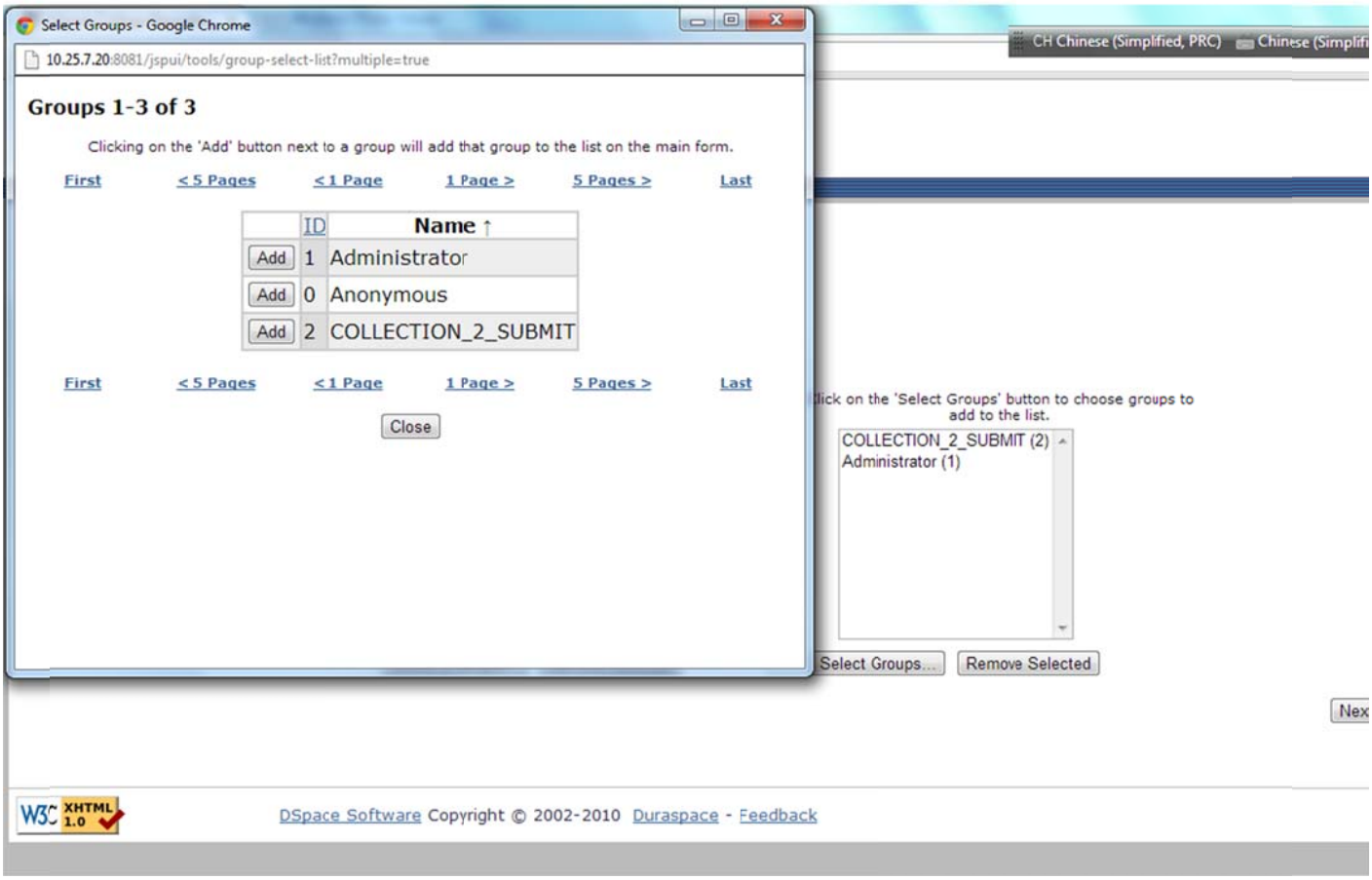
62. Select E-person and give authorization



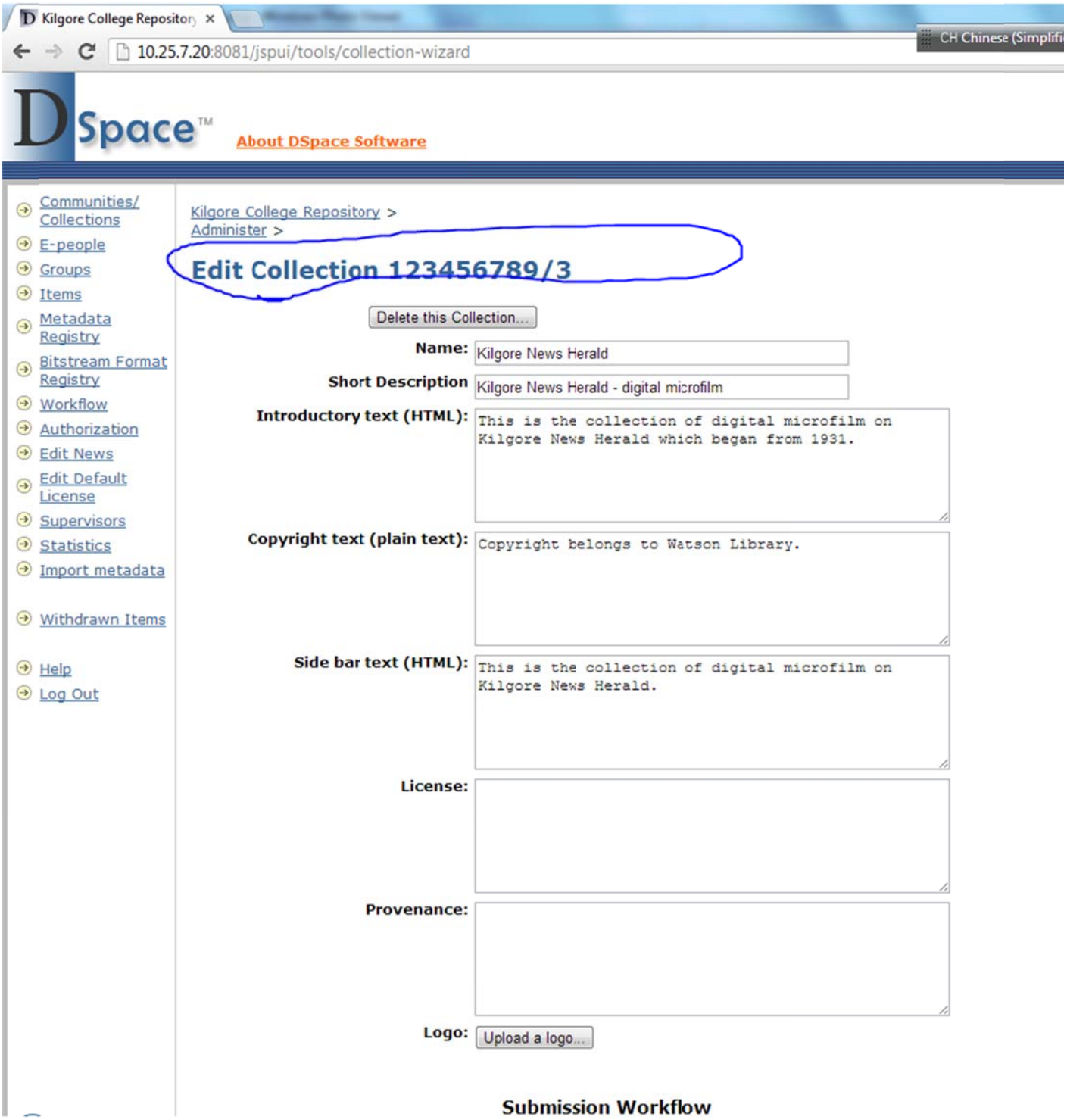
63. Add person



64. Add administrator and group



64. Collection number is allocated




The End